



MARKET DEMAND STUDY · COMMERCIAL PLANT SIZING

Mycoprotein Demand Study

How much mycoprotein the market will absorb per year — and the annual demand premise for sizing a commercial fermentation plant

A bottom-up estimate of annual mycoprotein demand (t/yr) across two application scenarios — human-supplement powder and meat-analogue wet biomass — in Brazil, the United States and Europe, under conservative, moderate and aggressive cases. Supported by competitive, pricing, regulatory, unit-economic and financial analysis that constrains the sizing answer — 35 exhibits across 19 sections.

Confidential — For conceptual plant design

Independent analysis · July 2026

Strictly private & confidential

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How to read this document

This is a decision document. Each section opens with the answer, then shows the evidence. The one-page thesis and scorecard are overleaf; the recommendation and phased playbook are in Section 19. Figures are triangulated from multiple third-party estimates and company disclosures and are presented as directional ranges — the alternative-protein data landscape is noisy, and where sources disagree we say so rather than manufacture false precision.

SECTION 00

Executive summary — the annual demand answer

This study estimates **how much mycoprotein the market will absorb per year**, in tonnes, so that the figure can serve as the sizing premise for the conceptual design of a commercial fermentation plant. Demand is built bottom-up from physical volumes across two application scenarios — mycoprotein as a **human-supplement protein powder** and as **wet biomass for meat-analogue products** — in Brazil, the United States and Europe, under conservative, moderate and aggressive cases. The remainder of the report exists to test and constrain that number: competitors, prices, regulation, unit economics and returns all bear on how much of the addressable market a new mycoprotein can realistically win, and at what price.

The answer, up front

Combining both scenarios on a common wet-biomass basis gives annual demand of roughly **4,700 t/yr (conservative)**, **16,900 t/yr (moderate)** and **41,100 t/yr (aggressive)**. The moderate case is the recommended planning premise, implying a first-plant nameplate of about **15,000–18,000 t/yr of wet biomass** ($\approx 4,200$ t/yr dry, $\approx 2,300$ t/yr protein), engineered for phased expansion toward $\sim 30\text{--}40\text{k}$ t/yr. Two constraints qualify it: a plant much below $\sim 10\text{k}$ t/yr is economically fragile (a 5k line needs 93% utilisation to break even), and the volume is only bankable if it is contracted — the returns analysis shows realised *price* matters as much as scale.

EXHIBIT 1

Estimated annual mycoprotein demand — the study's core deliverable

Case	Scenario 1 — supplement	Scenario 2 — meat analogue	Combined (wet biomass)	Read
Conservative	~ 750 t/yr powder	$\sim 2,000$ t/yr wet	$\sim 4,700$ t/yr	Below viable plant scale
Moderate	$\sim 3,000$ t/yr powder	$\sim 6,000$ t/yr wet	$\sim 16,900$ t/yr	Planning premise
Aggressive	$\sim 7,600$ t/yr powder	$\sim 14,000$ t/yr wet	$\sim 41,100$ t/yr	Expansion target

Source: Independent bottom-up demand model (Section 16). Scenario 1 is dry powder; Scenario 2 is wet biomass; the combined column converts both to wet-biomass equivalent at 1 t powder ≈ 3.6 t paste.

Confidence in the demand estimate

Addressable market data	SOLID	Supplement and plant-based-meat volumes triangulated from scanner and vendor data
Price benchmarks	SOLID	Competitor pricing available on a consistent protein basis
Scenario 1 demand (supplement)	MODERATE	Penetration of a novel ingredient is the least certain input
Scenario 2 demand (analogue)	MODERATE	Category is flat-to-declining; upside leans on hybrid and foodservice
Realistic share assumption	KEY UNCERTAINTY	A 1-point shift in penetration moves the nameplate materially
Conversion to physical tonnes	SOLID	Explicit dry / wet / product basis throughout
Price realisation at that volume	KEY UNCERTAINTY	15k plant needs $\sim \$3.19/\text{kg}$ wet for NPV = 0

Regulatory access to volume

MIXED

Europe already approved; US and Brazil navigable but not instant

SIZING PREMISE: ~15,000–18,000 t/yr wet biomass — moderate case, staged and contracted

Seven findings that shape the demand estimate

- **A small niche with a structural edge.** Mycoprotein is ~\$0.76B (finished product) to ~\$1.4B (broad ingredient) today — tiny next to the ~\$20B plant-protein pool and a rounding error against the broader meat market (a very wide reference that frames the ceiling, not the served market) — but it grows ~8–10% and carries a substantially lower footprint than animal proteins (and, depending on assumptions, competitive with plant proteins) while delivering complete protein plus prebiotic fibre.
- **The monetisation is shifting toward ingredients.** Functional ingredients — supplements, feed, binders and the base for hybrid products — are, on one broad-definition estimate, 70–80% of volume. That figure is soft (it sits awkwardly against Quorn’s finished-product dominance), but the direction is clear: mycoprotein is becoming an ingredient platform, not just a meat substitute.
- **Don’t confuse the plant-based slump with the death of alt-protein.** US retail meat-analogue sales are down ~19% since 2023, yet global plant-based meat still grew ~8% and capital is rotating hard into fermentation (~75% of Q1 2026 alt-protein funding). The failure was a branded-CPG, retail-shelf failure — not a failure of the protein source.
- **Business model is the whole game.** Survivors are B2B suppliers with feedstock integration and offtake (ENOUGH, backed by Cargill). Casualties are asset-heavy branded players: Meati raised ~\$450M and sold for ~\$4M; Mycorena went bankrupt; Quorn books heavy losses on ~£187M of revenue.
- **Attractive economics, one binding constraint.** Biomass cost of production is ~\$4–6/kg falling toward ~\$2.5/kg at ~32k-tonne scale — below beef and pork, far below precision fermentation — because it harvests whole biomass with light purification. (These figures must be normalised to a single basis — dry, protein, or product — before use; see Section 11.) The constraint is capital intensity (\$80–150M/plant) and the scale-up “valley of death.” Feedstock cost and RNA-reduction yield are the biggest levers.
- **The higher-value niches are specialised.** Beyond the commodity shelf sit **hybrid 50/50 meat** and differentiated **active-nutrition** formulations (\$15–25/kg), with **GLP-1 companion nutrition** a genuine but unproven option-value adjacency — promising given the muscle-loss dynamic, but not yet a demonstrated market and not a load-bearing part of the base case.
- **Regulation is moat and map.** The US (GRAS) is fastest, Singapore a quick beachhead, Brazil now navigable (ANVISA RDC 839/2023), and the EU slow (2–6 years) but credibility-conferring with a 5-year data-exclusivity reward. Recent tightening — an FDA deficiency letter (Feb 2026) and new UK genome-sequencing/DIAAS rules (Mar 2026) — raises the bar and favours the well-advised.

SECTION 01

Objective, scope and method

This study answers one question: **how much mycoprotein will the market absorb per year?** The output is an annual demand figure in tonnes that can be used directly as the sizing premise for the conceptual design of a commercial fermentation plant. Everything else in this report exists to make that number defensible — or to show where it breaks.

The two application scenarios

- **Scenario 1 — mycoprotein as a human-supplement ingredient.** Sold as a protein powder, alone or blended with other protein sources, into the food-supplement market. It competes with whey protein, soy protein isolate and pea protein; the stated differentiators are high protein content (>50 g/100 g) and a complete essential-amino-acid profile.
- **Scenario 2 — mycoprotein as a meat-analogue ingredient.** Wet biomass used as the principal ingredient in meat-alternative products, competing with plant-based products, with nutritional profile and texture as the differentiators.

The six questions answered for each scenario

- Which markets have the greatest consumption potential — **Brazil, the United States or Europe?** (noting that mycoprotein is already approved for sale in Europe)
- What is the **current size of that market and its growth rate?**
- Who are the **principal competitors and substitute products?**
- What **prices** are those products sold at today?
- What would be a **realistic market share** for a new mycoprotein?
- What is the **estimated potential demand under conservative, moderate and aggressive scenarios (t/yr)?**

Expected deliverable

The principal output of this study is an **estimated annual mycoprotein demand** that can be used as the premise for sizing the future commercial plant. Section 16 contains that estimate and the full derivation; Section 17 converts it into a plant nameplate and tests whether the resulting asset earns its cost of capital. Sections 02–15 supply the market, competitive, pricing, regulatory and cost evidence the estimate rests on.

Scope boundaries

- **Product.** Fungal biomass grown by fermentation for food — primarily mycoprotein in the strict sense (filamentous fungi such as *Fusarium venenatum*), plus adjacent fungal-biomass ingredients. Precision and gas fermentation appear only for contrast.
- **Geographies.** Brazil, the United States and Europe — the three markets named in the brief, which also carry the most relevant demand potential and the clearest (if uneven) regulatory routes.
- **Not in scope.** Detailed process design, equipment specification, site selection and engineering cost estimation. Where this report touches plant cost, it does so only to test whether the demand-derived capacity is economically coherent.

Method at a glance

We triangulated published market estimates, company filings and disclosures, non-profit industry data (Good Food Institute), retail/consumption data (SPINS, Euromonitor), peer-reviewed life-cycle and techno-economic studies, and regulatory registers, and we cross-checked the result against an independent parallel analysis. Where third-party

numbers diverge — and for market size they diverge by up to 5× — we present a range and a reasoned base case. All monetary figures are USD unless noted.

A note on data quality

Alternative-protein research is unusually noisy: definitions differ (does “mycoprotein” mean the ingredient, the finished product, or the whole meat-substitute shelf?), and vendors have incentives to publish large numbers. We flag the boundary for every figure and lean on primary disclosures — capacity, funding, filings — wherever possible. On sustainability in particular, headline multiples vs beef vary enormously with the baseline chosen; we report ranges, not single hero numbers.

SECTION 02

Answers to the study brief

The six questions posed for each scenario, answered in summary. Each answer is derived and evidenced in the sections that follow; the demand figures come from the bottom-up model in Section 16.

Scenario 1 — mycoprotein as a human-supplement ingredient

EXHIBIT 2

Scenario 1 — answers at a glance

Question	Answer
Greatest consumption potential	US is the largest market by value (~40% of a ~\$28–30B global supplement market) and the scale prize. Brazil is the largest South-American market and the natural home base. Europe is the fastest route to revenue because mycoprotein is already approved. Recommended sequence: Europe and Brazil first, US as the volume prize.
Current size and growth	Global protein supplements ~\$28–30B, growing ~6–10%/yr; powder is ~54% of that. The realistically addressable slice — plant-based / alternative-protein powder — is ~\$2.9B, equal to ~151,000 t/yr of powder across the three regions (US ~74k, Europe ~58k, Brazil ~19k).
Competitors and substitutes	Whey (Glanbia / Optimum Nutrition) is the entrenched benchmark; soy isolate (ADM); pea isolate (Roquette, Puris). Mycoprotein peers positioning powders: Marlow/Quorn ingredients, ENOUGH, The Protein Brewery.
Prices practised	Whey concentrate ~\$14.7/kg wholesale (~\$17/kg on a 100%-protein basis); retail whey \$25–32/kg; pea isolate \$15–25 retail (~\$5–9 bulk); soy isolate \$12–20 retail. A mycoprotein powder would target ~\$20–30/kg retail.
Realistic market share	0.5% (conservative) to 5% (aggressive) of the plant-protein-powder addressable segment — i.e. only ~0.1–1.3% of the total powder market. Mycoprotein is ~50–55% protein vs 80–90% for isolates, so it must sell on fibre and completeness, not grams per dollar.
Estimated demand (t/yr)	Conservative ~750 · Moderate ~3,000 · Aggressive ~7,600 t/yr of dry powder (≈ 2,700 / 10,900 / 27,200 t/yr expressed as wet biomass).

Source: Independent analysis; demand from the Section 16 model.

Scenario 2 — mycoprotein as a meat-analogue ingredient

EXHIBIT 3

Scenario 2 — answers at a glance

Question	Answer
Greatest consumption potential	Europe is the most attractive near term: largest regional market, already approved, and primed by Quorn familiarity. The US is the biggest single market but its retail segment is shrinking — enter via foodservice and hybrid, not branded retail. Brazil is smaller but growing ~11%/yr with existing distribution (Seara, Incrível).
Current size and outlook	Defensibly ~\$1.5B (US, retail-scanner-anchored; vendor all-channel figures run \$3–4B), ~\$2.7B (Europe), ~\$0.3B (Brazil) — together ~ 399,000 t/yr of finished product. Growth is contested: vendor CAGRs of 15–20% sit against flat-to-declining scanner volumes, so the physical base is treated as roughly stable.
Competitors	Beyond Meat, Impossible Foods and soy/pea incumbents globally. In Brazil: Fazenda Futuro (Future Farm), Marfrig (PicNic), BRF (Sadia Veg) and Seara Incrível. Quorn is the mycoprotein reference point.
Prices practised	Plant-based meat retails ~\$10–14/kg — roughly 2× beef and 4× chicken per kg (a ~77% premium in 2023, now narrowing). A mycoprotein-based product would target ~\$9–13/kg.
Realistic market share	1% (conservative) to 7% (aggressive) of the plant-based-meat market, reflecting genuine texture and nutrition advantages against cheaper, entrenched soy and pea incumbents.
Estimated demand (t/yr)	Conservative ~2,000 · Moderate ~6,000 · Aggressive ~14,000 t/yr of wet biomass (product volume × ~50% paste content).

Source: Independent analysis; demand from the Section 16 model.

The consolidated answer

Converting both scenarios to a common wet-biomass basis and summing them: **~4,700 t/yr (conservative), ~16,900 t/yr (moderate), ~41,100 t/yr (aggressive)**. The moderate case is the recommended premise for conceptual design, giving a first-plant nameplate of about **15,000–18,000 t/yr of wet biomass** with expansion headroom to ~30–40k. Section 16 derives these figures; Section 17 tests whether a plant of that size is economically coherent.

SECTION 03

What mycoprotein is — and where it sits

Mycoprotein is protein-rich fungal biomass grown by fermentation and harvested whole. Quorn has grown *Fusarium venenatum* in steel tanks since the 1980s; a wave of newer entrants now grows other fungi by the same basic route. Understanding the fermentation taxonomy is essential, because the four approaches have **radically different economics** — and most of the investment noise (and most of the failures) sits outside biomass fermentation.

EXHIBIT 4

Four fermentation routes — mycoprotein is a biomass play

The economics differ by an order of magnitude across these rows.

Route	What it does	Representative players	Mycoprotein?
Traditional fermentation	Microbes transform a substrate (age/culture food)	Tempeh, koji, cultured dairy	Adjacent — not the focus
Biomass fermentation	Grow whole microbial cells fast; harvest the biomass itself	Quorn, ENOUGH/ABUNDA, Nature's Fynd, Meati, Fermotein, Rhiza, PEKILLO	YES — mycoprotein lives here
Precision fermentation	Engineer microbes to secrete one specific molecule	Perfect Day (whey), Remilk (casein), Impossible (heme)	No — costlier, purification-heavy
Gas fermentation	Feed microbes CO ₂ / H ₂ / methane instead of sugar	Solar Foods (Solein), Calysta, Unibio	No — earlier-stage

Source: Independent analysis; company disclosures.

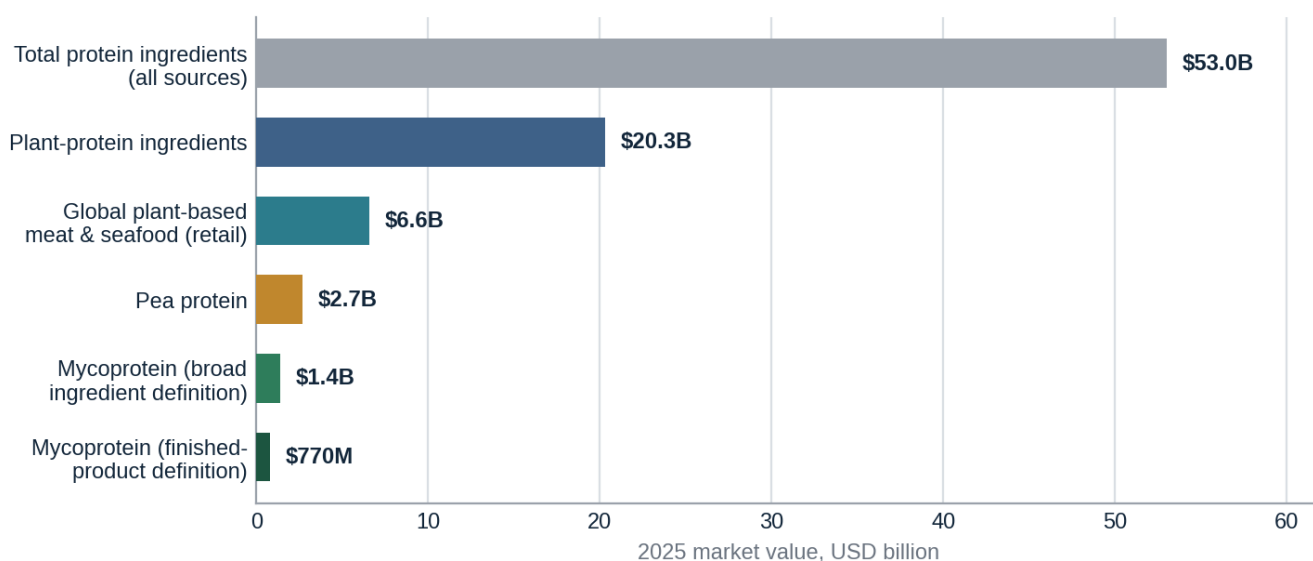
Why biomass fermentation is the sweet spot

Because biomass fermentation grows and harvests the *entire* cell rather than isolating and purifying a single molecule, its downstream processing is far lighter than precision fermentation's. That is the root of its cost advantage — and the reason mycoprotein can credibly chase parity with animal protein while precision fermentation still struggles against commodity ingredients. It also yields a whole-food ingredient (protein + fibre) rather than an isolate, which shapes both its nutritional positioning and its consumer story.

EXHIBIT 5

Mycoprotein is a sliver of the protein economy today

Where the category sits within the nested protein-market stack (2025).



Source: Independent analysis triangulating published estimates; GFI; Euromonitor. Boundaries differ by source; figures are directional.

The strategic read

Being small is the opportunity, not the problem. Mycoprotein is early enough that a single well-run 20k-tonne plant is a *material* share of global supply — unlike pea protein, where a new entrant is a price-taker against entrenched majors. The question is not “is the market big?” but “can we build a low-cost, reliable supply position before the category scales?”

SECTION 04

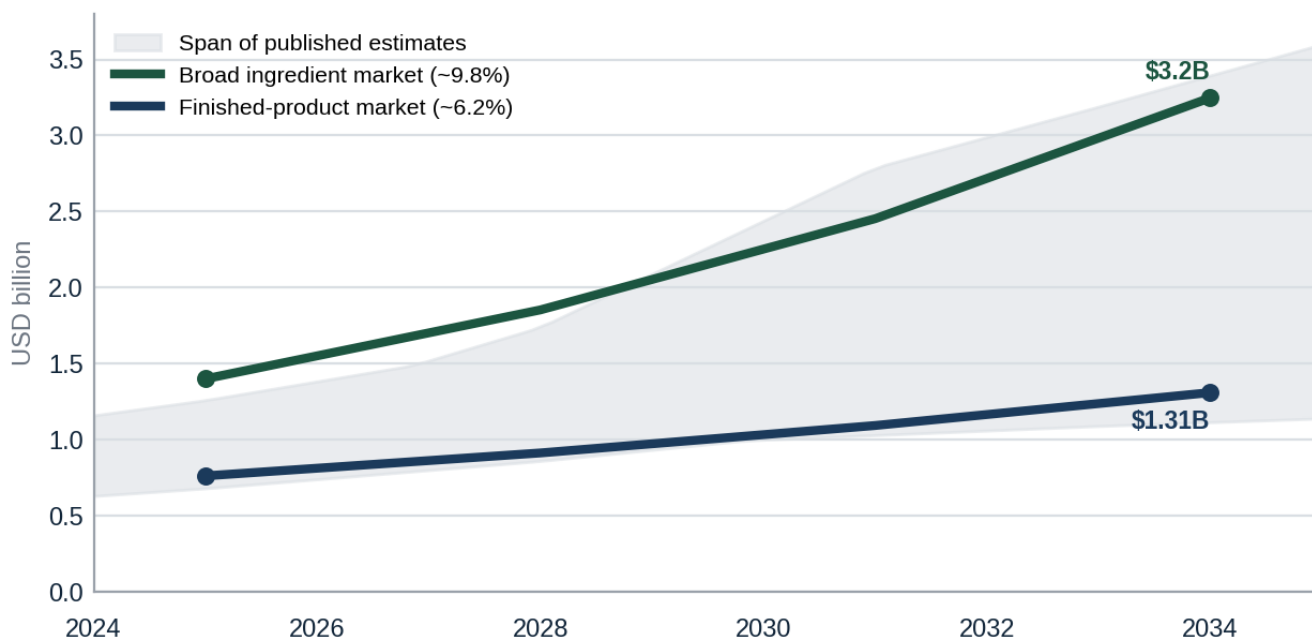
Market size and growth

The number you quote depends entirely on where you draw the boundary. The **finished-product** market is ~\$0.76B in 2025, growing ~6.2% to ~\$1.4B by the mid-2030s. The **broad ingredient** market — including supplements, feed and functional additives — is ~\$1.4B in 2025, growing ~9.8% to ~\$3.0B by 2034. Published estimates span an even wider cone. That spread is itself a finding: the trajectory hinges on whether the category escapes the branded-retail model dragging on the wider meat-analogue shelf.

EXHIBIT 6

Two defensible market definitions, one wide forecast cone

Global mycoprotein value; shaded band spans published third-party estimates.



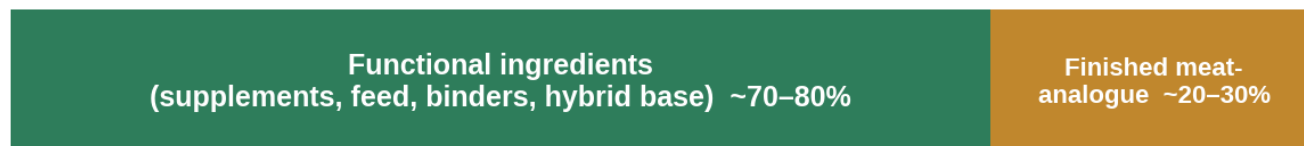
Source: Independent analysis; band reflects five published estimate sets; two headline trajectories reflect finished-product vs broad-ingredient boundaries.

The monetisation has shifted toward functional ingredients

EXHIBIT 7

The monetisation has shifted toward functional ingredients

Approximate split of mycoprotein consumption by use.



Single broad-definition commercial estimate — directional only; hard to reconcile with Quorn's finished-product dominance

Source: Independent analysis; industry estimates.

The takeaway for an entrant: **value is migrating from finished meat-analogues to ingredients** — supplements, hybrid bases and binders — which is exactly the higher-margin, lower-marketing terrain a B2B model should target.

Put in context against the comparison markets

- **vs plant-protein ingredients (~\$20B, ~8.5% CAGR).** Mycoprotein is tiny beside the pea/soy/wheat pool but growing at a similar-or-faster clip from a far smaller base — the classic profile of an emerging ingredient.
- **vs pea protein (~\$2.7B).** Even plant protein's current darling is only ~2–3.5× mycoprotein's size; mycoprotein's differentiation (fibre, complete EAAs, sustainability) is arguably sharper.
- **vs global plant-based meat & seafood (~\$6.6B retail).** Mycoprotein's most direct end-market; it grew ~8% globally in 2025 even as the US retail slice fell — demand is real but geographically uneven.
- **vs the global meat market (~\$2.3T).** Worth stating only to be dismissed: mycoprotein does not compete for the whole meat value chain, and a TAM that wide is not a planning number. The relevant served market is the plant-based and alternative-protein pool sized in Section 16, not this.

SECTION 05

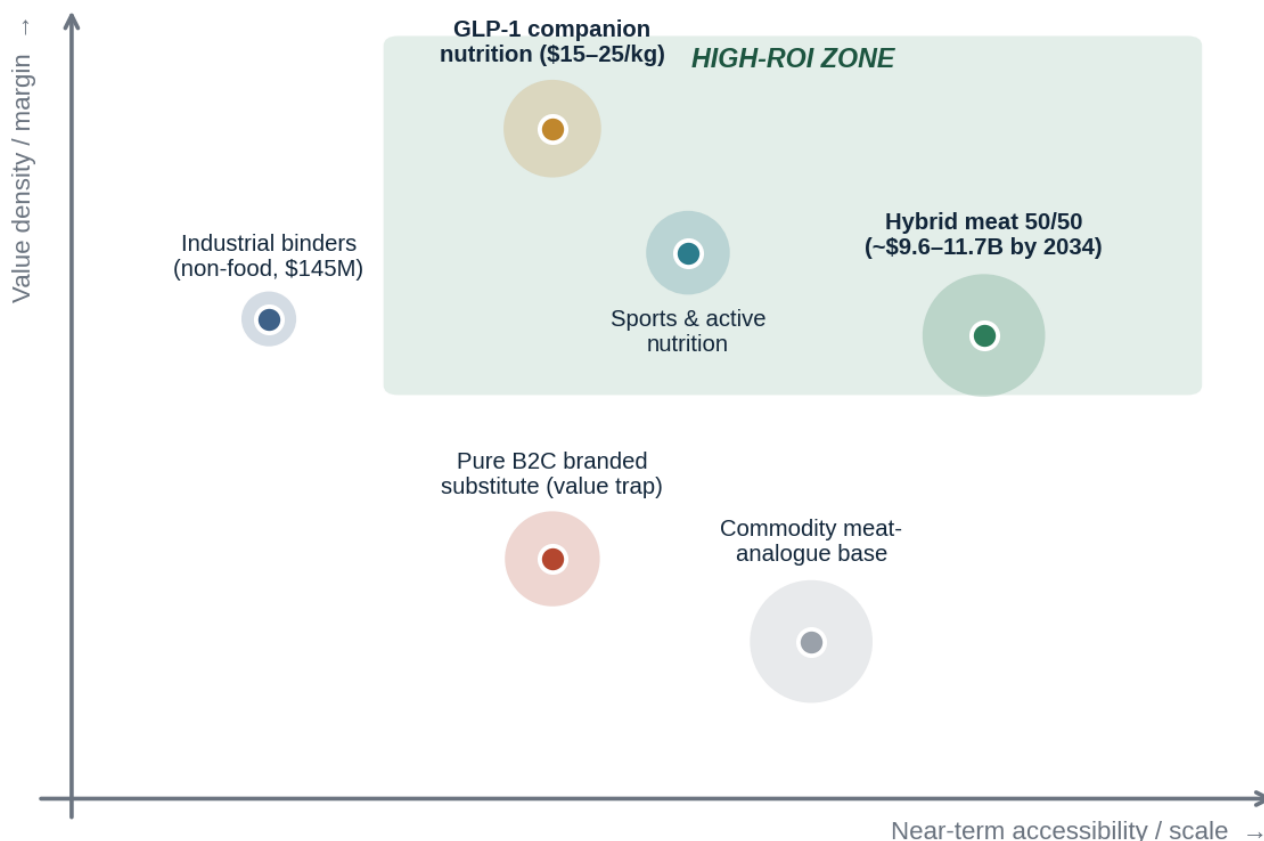
Where the value is moving — the niche map

The most viable paths to profit deliberately depart from the original dogma of replacing meat with 100% plant replicas. Three niches stand out, and two of them are where a disciplined entrant should concentrate.

EXHIBIT 8

Play the high-ROI niches, not the commodity shelf

Opportunity map by near-term accessibility and value density; bubble size $\approx$ market size.



Source: Independent analysis; market sizes from published estimates (hybrid meat ~\$9.6–11.7B by the mid-2030s — a broad, all-inclusive hybrid TAM; mycelium binders ~\$145M by 2036).

Hybrid meat — the pragmatic volume play (with a caveat)

Blending mycoprotein with conventional meat (e.g. 50/50) cuts carbon footprint, saturated fat and cost while preserving the familiar taste of animal fat and adding fibre, and it is sold to committed meat-eaters rather than vegans. Broad forecasts put the hybrid-meat market at **~\$9.6–11.7B by the mid-2030s** — but that is a wide, all-inclusive “hybrid” TAM (spanning plant, cultivated and insect blends), not mycoprotein-specific, so treat it as directional. And the neophobia advantage is not automatic: some consumer research finds tested hybrid products rated *below* both beef and plant-based, so hybrids need real commercial trials, not assumption. Quorn's foodservice UltiMeat range (UK, 2025/26) is an encouraging proof point, not proof of a validated volume market.

GLP-1 companion nutrition — a high-potential adjacency, not a proven market

GLP-1: promising option value — handle with care

GLP-1 receptor agonists (Ozempic and successors) suppress appetite and can cause substantial loss of lean mass — in some patients a reported **25–40% of weight lost** (lean mass, which is not solely skeletal muscle). That plausibly creates demand for nutrient-dense, high-protein, high-fibre foods in small portions, which fits mycoprotein's profile, and premium formulations can command **\$15–25/kg**. But this is a *hypothesis*, not a demonstrated market: there is no established sales base, no proven willingness-to-pay, and stiff competition from whey (the clinical gold standard for muscle protein synthesis), dairy and amino-acid products. The Novo Holdings investment in The Protein Brewery signals strategic interest — it does not validate commercial demand. Treat GLP-1 as **option-value upside, not a base-case pillar**.

Industrial binders — optionality outside food

Beyond food entirely, fungal *mycelium materials* (a distinct technology, product set and competitor field from food mycoprotein) are being explored as sustainable binders in engineered wood, packaging and insulation — a replacement for formaldehyde resins. The addressable market is small (~\$145M by 2036) but sits *outside* food-safety regulation, offering a lower-friction outlet and a use for off-spec biomass. It is an adjacency to watch, not a core plank.

SECTION 06

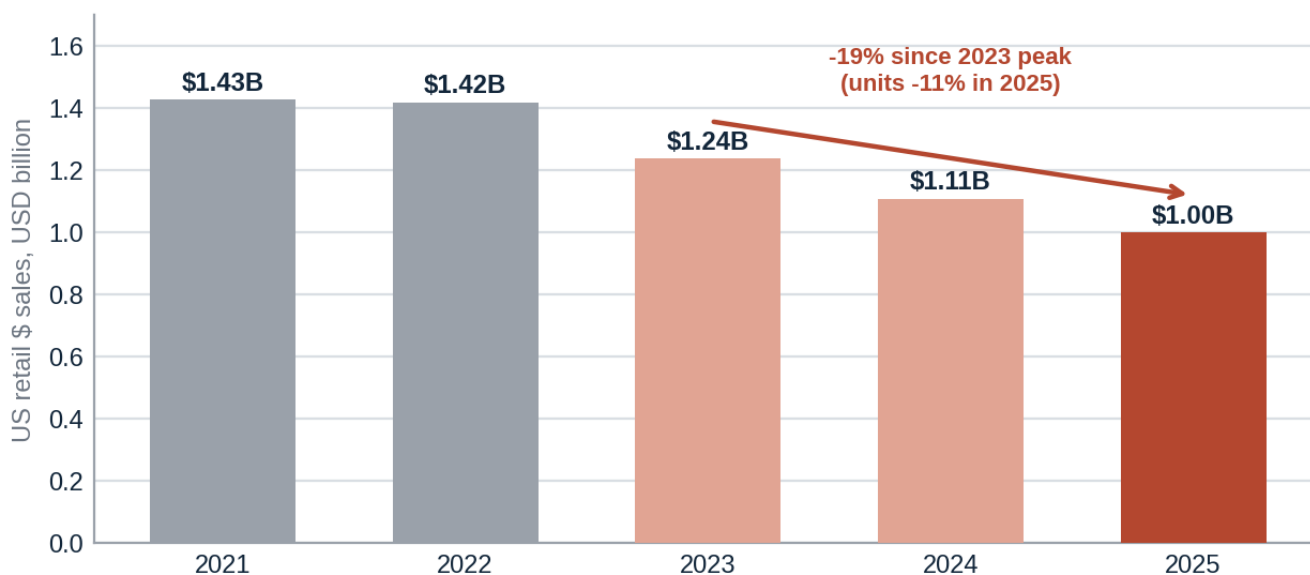
The plant-based reality check

Any entry thesis must confront the elephant in the room: branded plant-based meat has been in a multi-year slump in its most developed markets. This is the most important context for an investor — and the most misread. **The slump is a business-model and product problem, not proof that consumers have abandoned protein diversification.**

EXHIBIT 9

US retail alt-meat has fallen for three straight years

US retail dollar sales of plant-based meat & seafood (illustrative, triangulated).

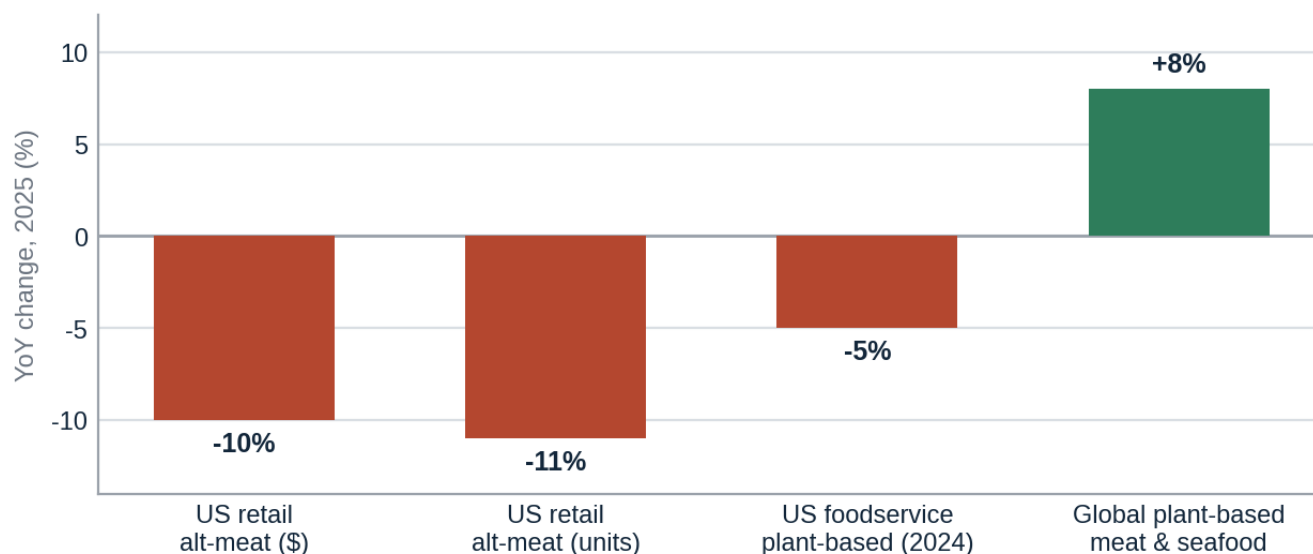


Source: Independent analysis anchored to GFI/SPINS data points (2025 ~\$1.0B, -10% YoY, units -11%; -19% since 2023). Annual path is illustrative.

The causes are well understood: premium pricing into an inflation-squeezed consumer, taste and texture that under-delivered, a growing “ultra-processed” perception, and over-distribution that outran genuine repeat demand. But the category is *repositioning*, not uniformly dying.

EXHIBIT 10**Not dying everywhere — repositioning**

Divergent 2025 growth signals across channels and geographies.



Source: Independent analysis; GFI; SPINS; Euromonitor. US foodservice figure is 2024.

Plant protein still leads the substitute shelf — but its constraints are showing**EXHIBIT 11****Plant protein still leads the substitute shelf**

Approximate share of the meat-substitute category.



Source: Independent analysis; industry estimates.

Why this helps a disciplined mycoprotein entrant

First, the reset has cleared weak competitors and cheapened assets and talent. Second, it validates the pivot the smart money is already making — away from own-brand retail toward **B2B ingredient supply, foodservice and hybrid products**, where mycoprotein's texture and nutrition actually win. An entrant who never builds a consumer brand sidesteps the exact trap that snared the category's pioneers.

SECTION 07

Demand drivers and the consumer landscape

Underneath the retail noise, the structural demand case is robust — but consumer-facing adoption faces real cultural friction that shapes *where* and *how* to sell.

The structural drivers

- **Sustainability as risk mitigation.** Animal agriculture drives ~a third of food-system GHG emissions and ~70% of freshwater use. As Scope-3 commitments proliferate among majors like Nestlé and Unilever (these vary by firm and are rarely a direct legal obligation), lower-footprint proteins increasingly become a supply-chain risk-management lever, not just a marketing choice.
- **Health megatrends.** The WHO classifies processed red meat as Group-1 carcinogenic; the “flexitarian” cohort — reducing meat without full vegetarianism — is now ~42% of adults in key markets, and is highly taste- and texture-sensitive, favouring mycoprotein’s natural fibrousness.
- **The GLP-1 wave.** As covered in Section 4, appetite suppression plus muscle loss is generating unprecedented demand for nutrient-dense high-protein foods.

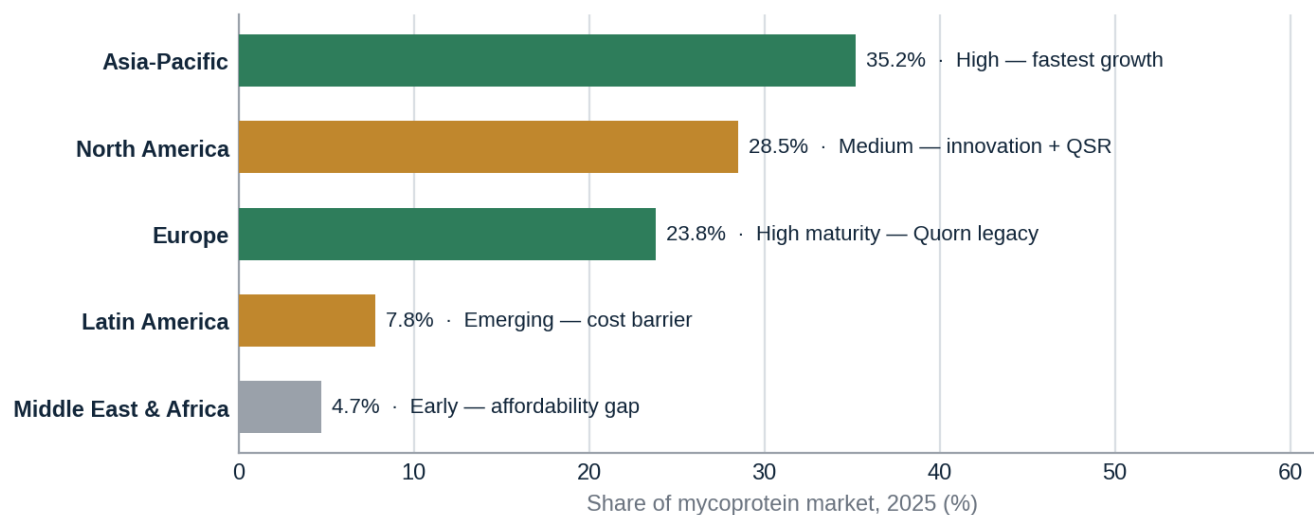
The consumer barriers — and the regional map

The core friction is semantic: “fungus,” “biomass” and “microbe” trigger visceral food neophobia in many Western consumers (only ~16% report knowingly eating “fungus meat”), pushing brands toward opaque naming like “fermented protein.” Acceptance varies sharply by region — which should shape market sequencing.

EXHIBIT 12

Acceptance and market share vary sharply by region

Mycoprotein market share and adoption posture by region (2025).



Source: Independent analysis; published regional estimates.

SECTION 08

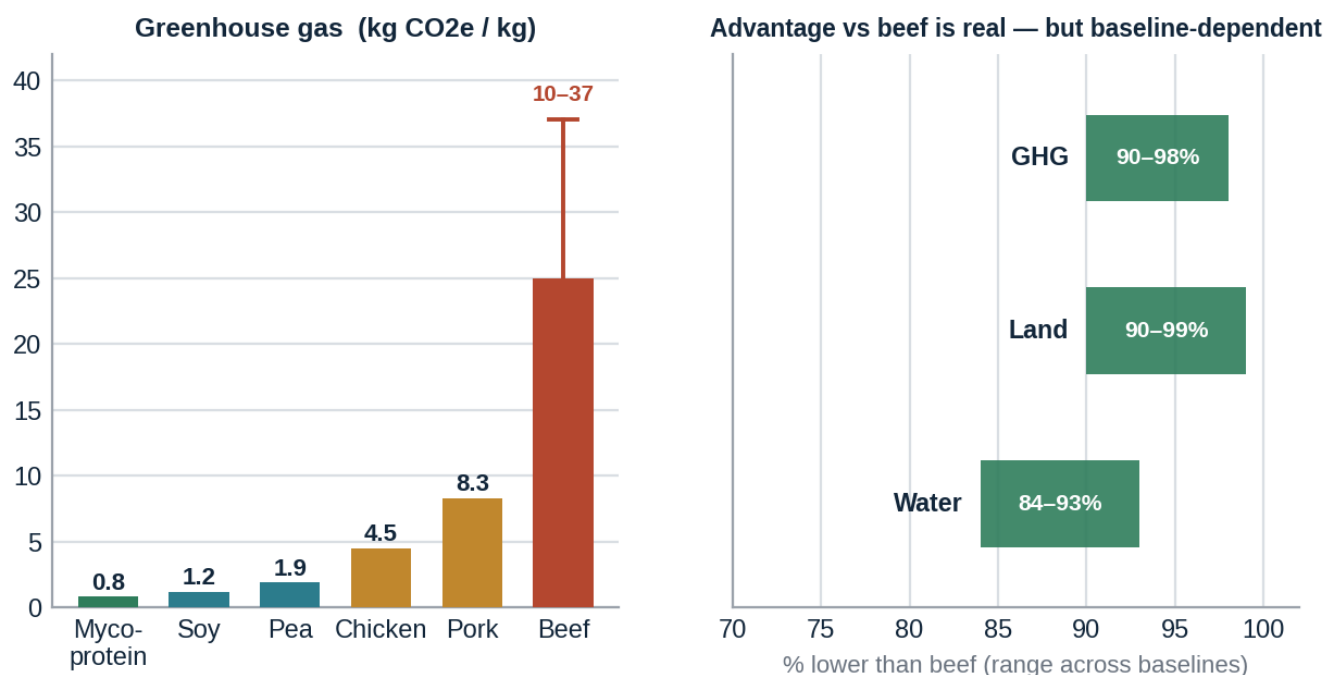
Why mycoprotein is structurally different

Mycoprotein is not just “another plant-based option.” On the two axes investors should care about most — **sustainability** and **nutrition** — it holds a clear, evidence-backed edge over animal proteins and, depending on assumptions, a competitive position versus plant proteins. That edge is a durable reason the category outlives the current downturn.

EXHIBIT 13

Lower footprint than meat — and than plant proteins

Cradle-to-gate impacts, and the (baseline-dependent) advantage over beef.



Source: Peer-reviewed LCAs and systematic reviews; producer footprint reports. Beef GHG spans 10–37; advantage vs beef varies with the baseline chosen.

The honest sustainability headline

Mycoprotein emits roughly **0.7–0.8 kg CO₂e/kg** — far lower than any meat, and often lower than soy and pea protein, though the margin over *plant* proteins is genuinely conditional (it depends on the electricity mix, the glucose feedstock’s own footprint, protein basis and co-product allocation, and can narrow or invert under some boundaries). Its advantage over beef is large but *baseline-dependent*: peer-reviewed reviews put it near **90% lower**, producer LCAs against grass-fed beef reach **~98%**. The defensible framing for a data room: **dramatically lower than animal proteins; competitive with plant proteins depending on assumptions.**

Nutrition — strong, but be honest about protein density

Mycoprotein is a complete protein (all nine essential amino acids) with a **PDCAAS reported near ~0.99** for *Fusarium venenatum* (Quorn) — approaching milk. An important caveat: that figure is organism- and process-specific, not a universal “mycoprotein” constant — the different species this report covers (*F. venenatum*, PEKIL0/*Paecilomyces*, others) differ in protein content, fibre, RNA and digestibility, so nutritional specs should be validated per strain. The category still avoids the antinutrients (phytates, lectins) and amino-acid gaps (e.g. pea’s methionine, wheat’s lysine) that dog some plant proteins, and its natural hyphal structure recreates meat’s bite without the harsh high-moisture extrusion

that globular plant proteins require. But on *raw protein density* the isolates win — whey/soy/pea deliver ~80–90 g/100g versus mycoprotein's ~45–50 g. The supplement play must therefore be a differentiated *protein-plus-fibre* proposition, not a head-to-head on grams per dollar.

EXHIBIT 14**Whole-food strength; isolate-grade density is not the pitch**

Attribute	Mycoprotein	Pea isolate	Soy isolate	Whey
Protein (per 100 g, dry)	~45–50 g	~80 g	~85–90 g	~80 g
Dietary fibre	High (prebiotic)	Low	Low	None
Complete amino acids	Yes (all 9)	Near (low Met)	Yes	Yes
PDCAAS	~0.996	~0.89	~0.91	~1.00
Antinutrients	None notable	Phytates	Phytates	None
Processing to fibre	Natural hyphae	HME extrusion	HME extrusion	n/a (animal)
Footprint vs peers	Lowest	Low	Low	High

Source: Independent analysis of typical published compositions; values vary by product and preparation.

SECTION 09

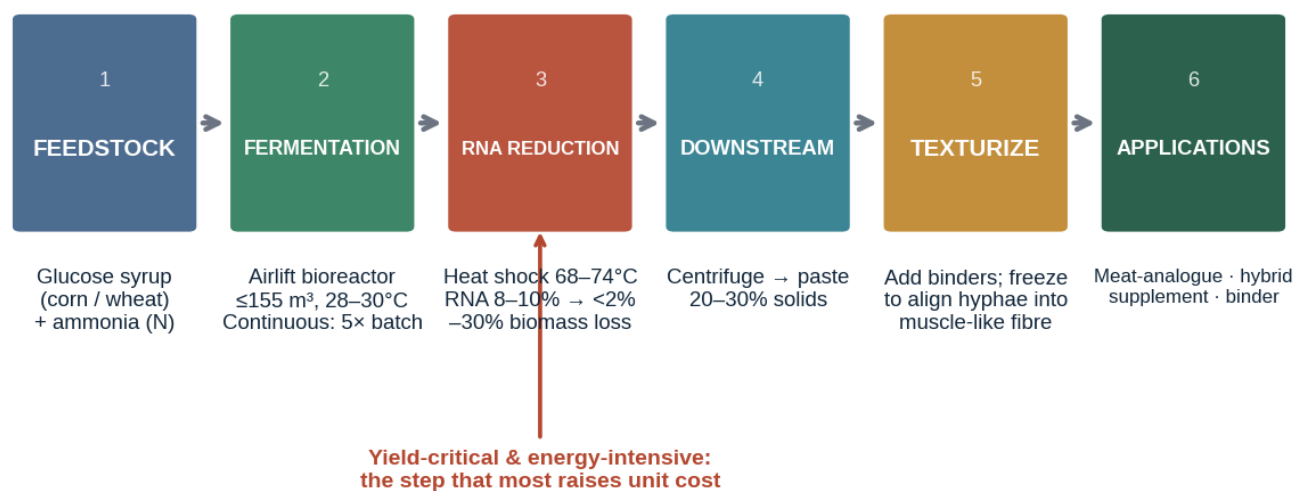
The technology and value chain

Mycoprotein's economics are decided upstream, in a handful of process steps. Understanding them is not academic — two of these steps (RNA reduction and strain stability) are where unit cost and operational risk actually live, and they explain why the category is capital- and know-how-intensive.

EXHIBIT 15

From sugar to muscle-like fibre — and where the cost hides

Simplified biomass-fermentation value chain with key parameters.



Source: Independent analysis; process parameters from published mycoprotein literature and patents.

The two steps that decide the economics

- **RNA reduction is the yield killer.** Fast-growing fungal biomass contains 8–10% RNA; consumed in excess it raises uric acid (gout, kidney stones), so the broth is heat-shocked (68–74°C, 30–45 min) to activate RNases and cut nucleic-acid content. The commonly-cited target — nucleic acid under ~2% of dry weight, tied to a ~2 g/day intake ceiling — derives from older UN PAG/FAO-WHO protein guidance (a processing/safety target, not a single crisp modern statute; verify the exact reference and basis per market). This step is energy-hungry and destroys up to **30% of the final solid biomass** — a direct hit to unit economics. (The released nucleotides carry umami value and can be sold as flavour enhancers.)
- **Strain mutation forces costly stoppages.** In continuous culture, a branched *Fusarium* mutant tends to emerge and out-compete the parent after ~100 generations (~6 weeks), degrading the linear-filament structure that makes the meat-like texture — forcing a shutdown, sterilisation and re-inoculation cycle. Managing this is core operational know-how.
- **Continuous fermentation is a requirement, not a nicety.** It delivers up to **5×** the productivity of batch operation and is essential to approach parity with conventional meat; airlift bioreactors up to 155 m³ run at 28–30°C under tight pH and oxygen control.
- **Feedstock and formulation.** Glucose syrup (corn/wheat) plus ammonia is the base; the centrifuged paste (20–30% solids) is bound (egg albumin for vegetarian; potato protein, gluten or alginate for vegan) and frozen to align hyphae into fibre. The circular-economy frontier — using agro-industrial sidestreams as feedstock (e.g. Enifer's PEKILLO) — both cuts opex and strengthens the ESG story.

SECTION 10

The competitive landscape

The field splits into three tiers by production maturity and strategy. The most important competitive fact for an entrant: the **B2B, commodity-scale corner is still sparsely populated** — ENOUGH is effectively alone at scale, with Enifer scaling behind it.

EXHIBIT 16

The mycoprotein field: who's who

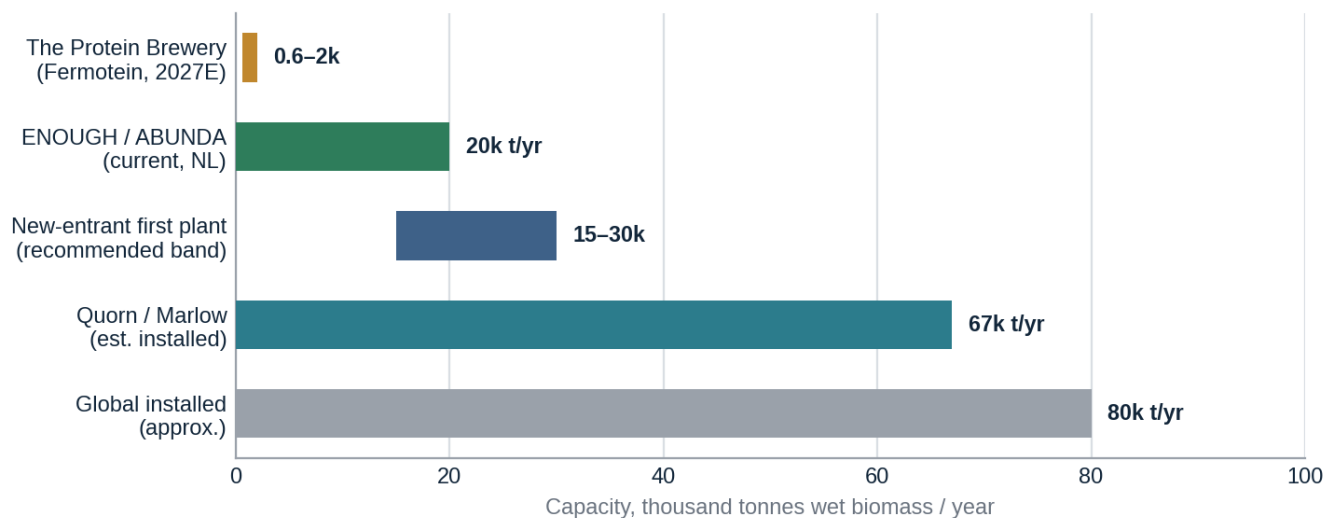
Player	Organism & process	Go-to-market	Capacity	Ownership & funding	Status / notes
Quorn / Marlow (Monde Nissin)	F. venenatum; biomass	Branded CPG + nascent B2B	~67k (est.)	Acq. £550M (2015); heavy losses	Leader ~24–31% global; £187M rev (2024, -9%); QuornPro + hybrids
ENOUGH / ABUNDA	F. venenatum; biomass	B2B ingredient	20k (site ≥60k)	~€95M+; Cargill stake + offtake	~13–15% share; co-located with Cargill; Unilever, M&S;
Nature's Fynd	F. flavelloides (Fy); surface ferment.	Branded + ingredient	n/d	\$500M+ (Gates, Bezos, Gen. IM)	US GRAS 2021; premium; capital-heavy
Meati (Emergy)	Mushroom mycelium; submerged	Branded whole-cut	"mega" plant (CO)	~\$450M → sold ~\$4M	Cautionary tale: covenant sweep → distress sale → revival
The Protein Brewery	R. pusillus (Fermotein)	B2B ingredient	0.6k → 2k (2027–29)	~\$60M; Novo Holdings	First novel mycelium approved in EU (2026); active-nutrition focus
Enifer	PEKILO; circular fermentation	B2B ingredient	Scaling 2026	~€49M; €33M plant	Agro-sidestream feedstock; Kantvik (FI); pet food + human
Also / distressed	Various fungi	Mixed	Early / n/a	Sub-scale	Mycorena (bankrupt → Naplasol); Better Meat (Rhiza, GRAS 2024); MyForest

Source: Company disclosures, filings and press; independent analysis. "n/d" = not disclosed. Capacities are wet-biomass estimates, t/yr.

EXHIBIT 17

Sizing intuition: where a first plant should land

Installed / target capacity benchmarks, thousand tonnes wet biomass per year.



Source: Company disclosures; independent analysis. Global installed ~80k is an approximation.

EXHIBIT 18**The attractive white space is B2B at commodity scale**

Strategic map by go-to-market and positioning; bubble size ≈ scale/funding.



Source: Independent analysis.

How to win

Play where ENOUGH and Enifer play — **B2B ingredient supply at commodity scale** — and out-execute on the two things that decide this market: **delivered cost** (feedstock integration) and **application performance** (win hybrid and poultry-style uses where mycoprotein's fibre and texture are unmatched). Brand is not the battleground; reliable, low-cost, spec-consistent supply is.

SECTION 11

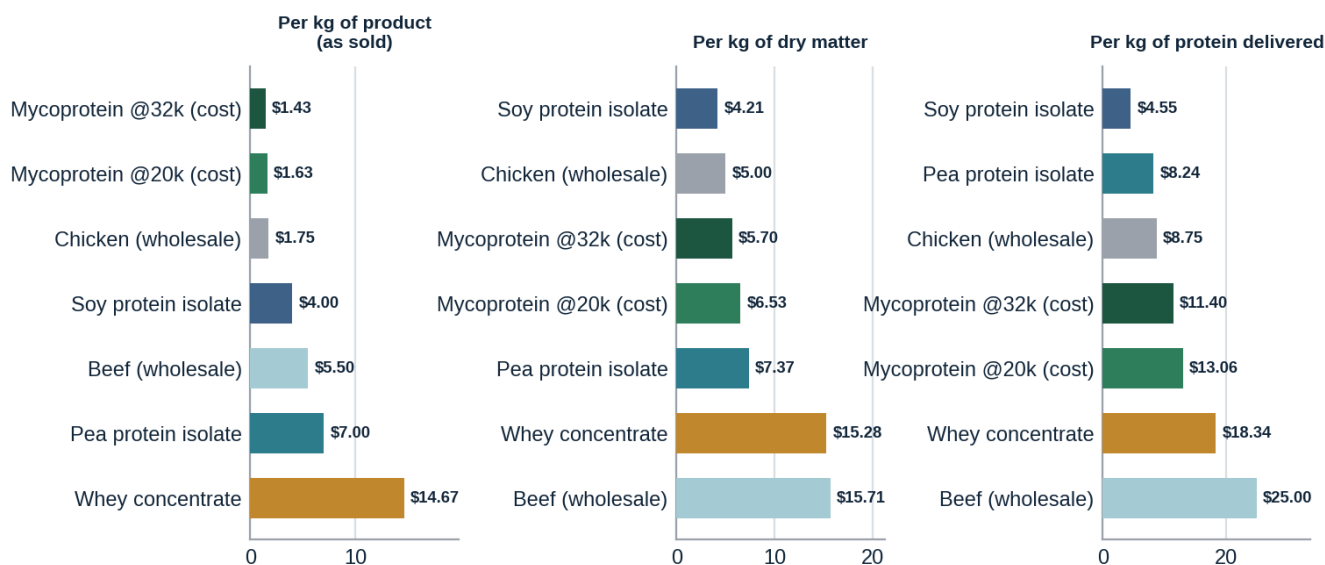
Value chain and unit economics

Here is the good news the consumer struggles obscure: **biomass fermentation has credible unit economics**. But the published shorthand — “\$4–6/kg falling to ~\$2.50/kg” — is unusable until it is pinned to a basis, because mycoprotein is sold wet (~25% solids) while meat is sold as product and isolates are sold nearly dry. Comparing them without normalising flatters the economics. Below we rebuild the comparison on **three separate bases**, then show what a plant actually costs at each candidate scale.

EXHIBIT 19

The same comparison on three bases — and it changes the answer

Cost or wholesale price per kg of product sold, per kg of dry matter, and per kg of protein delivered.



Source: Independent bottom-up model (mycoprotein, full cost at 90% utilisation); commodity and ingredient price ranges from published market data. Protein and dry-matter fractions stated in Appendix A.

What normalisation does to the thesis

On a **per-kg-of-product** basis mycoprotein paste looks cheap (~\$1.60/kg) — but that is mostly water. On a **per-kg-of-protein** basis, the honest comparison, modelled mycoprotein at 20k t/yr lands near **\$13/kg protein**: comfortably below beef (~\$25) and whey (~\$18), but *above* chicken (~\$9) and above pea and soy isolate (~\$8 and ~\$4.5). The earlier claim that mycoprotein “beats beef and pork” survives; the implication that it is broadly cost-advantaged against plant proteins does not. The commercial case must therefore rest on **differentiation** — fibre, texture, complete amino acids, footprint — not on being the cheapest gram of protein. Note also that these are ex-works figures: processing, formulation, packaging and logistics still sit between this and a finished product on shelf.

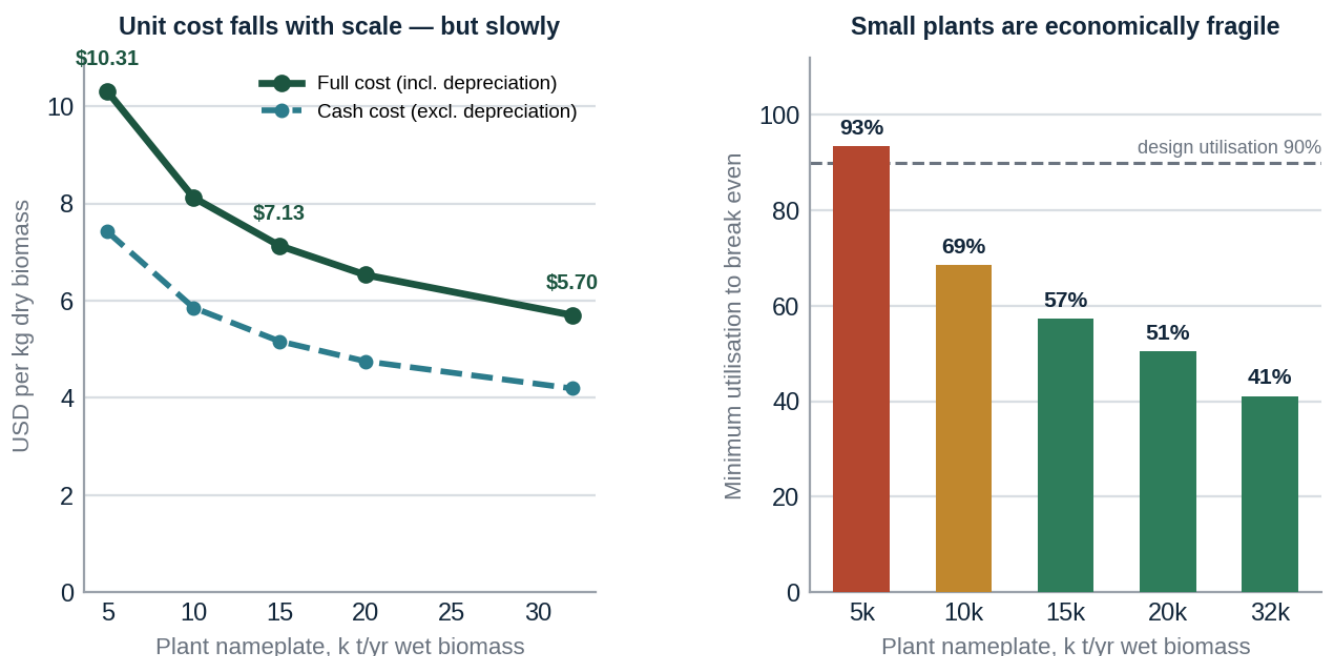
What a plant costs at each candidate scale

The single most consequential gap in a capacity decision is the difference between the cost quoted at *full* scale and the cost actually achieved at *entry* scale. Modelling capex with a capacity-scaling exponent and fixed opex sub-linearly gives the curve below — and with it the minimum utilisation each plant size must sustain simply to break even.

EXHIBIT 20

Scale economics — and why very small plants are fragile

Left: modelled unit cost by nameplate. Right: minimum utilisation to cover cash cost plus depreciation.



Source: Independent bottom-up model; assumptions and scaling exponents in Appendix A.

EXHIBIT 21

Cost and minimum utilisation by plant size

Nameplate (wet)	Capex	\$/kg dry	\$/kg wet	\$/kg protein	Break-even utilisation
5,000 t/yr	~\$49M	\$10.31	\$2.58	\$20.61	93% — fragile
10,000 t/yr	~\$76M	\$8.12	\$2.03	\$16.24	69%
15,000 t/yr	~\$100M	\$7.13	\$1.78	\$14.26	57%
20,000 t/yr	~\$120M	\$6.53	\$1.63	\$13.06	51%
32,000 t/yr	~\$163M	\$5.70	\$1.43	\$11.40	41%

Source: Independent bottom-up model at 90% utilisation; full cost includes depreciation over a 15-year life.

Reconciling the published benchmark with this model

Techno-economic meta-analyses cluster fungal-biomass cost around **\$4–6/kg**, with private benchmarks near **\$2.50/kg at ~32k t/yr**. This model returns **\$5.70/kg dry** (full cost) or **\$1.43/kg wet** at that same scale — which straddles rather than matches the benchmark, because the published figure is rarely tagged to a moisture basis. The gap is not a discrepancy to be argued away: it is the single most important thing to resolve with the licensor or engineering contractor, since a benchmark quoted on a dry basis implies this model is ~2× conservative, while one quoted on a wet basis implies it is optimistic. Until that is pinned down, treat the cost curve as a *structure* — the shape of scale economics and the break-even utilisations are robust; the absolute level is not.

The finding that should govern the sizing decision

A **5,000 t/yr** line must run at **93% utilisation** just to break even — there is almost no margin for a slow commercial ramp, and it is the reason a “small and safe” first plant is not actually safe. By **15,000 t/yr** the break-even falls to ~57%, which a three-year ramp can clear. This is the quantitative case for the demand-derived nameplate in Section 16 — and against commissioning a token first line unless it is explicitly a pilot funded as R&D rather than as a commercial asset.

The binding constraint is capital, not cost-per-kilo

The reason mycoprotein has not scaled faster is not that a kilogram is expensive to make — it is that **getting to first commercial kilograms is capital-intensive and operationally hard**. A greenfield plant for 10–20k t/yr runs **\$80–150M** — which on these figures implies roughly **\$4,000–15,000 per annual tonne** of capacity; the fermenter itself is only ~30% of that, with downstream processing another ~35%. More than 60% of biomanufacturing ventures that succeed in the lab never reach commercial scale — the “valley of death” that took Meati despite \$450M of funding.

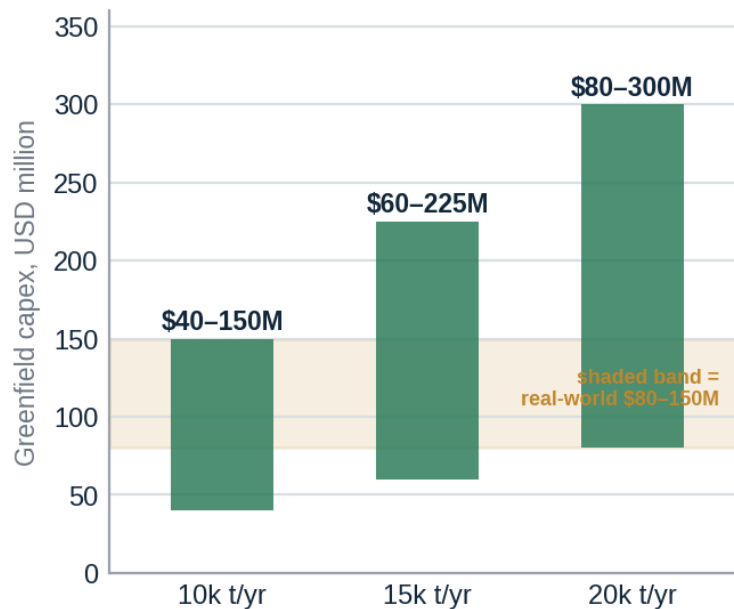
EXHIBIT 22

Capex structure and the plant-size trade-off

Left: typical capex composition. Right: greenfield capex by plant size.

Fermentation plant capex mix

- Downstream processing 35%
- Fermenter & seed train 30%
- Utilities & infrastructure 20%
- Building & other 15%

Capex vs plant size (~\$4–15k / annual tonne)

Source: Published bioprocess TEA rules of thumb; independent analysis.

The one lever that matters most: feedstock

Feedstock (fermentable carbohydrate) is ~a third of cash cost, and the models are clear that viability needs a carbon source below ~\$0.40/kg and high conversion yield. This is **exactly** why ENOUGH co-located with a Cargill starch plant — taking cheap sugars in and sending waste streams to an adjacent bioethanol unit in a zero-waste loop, sharing even waste process heat. Any entrant with privileged access to low-cost carbohydrate and industrial utilities starts with a structural margin advantage a pure-play startup cannot replicate.

SECTION 12

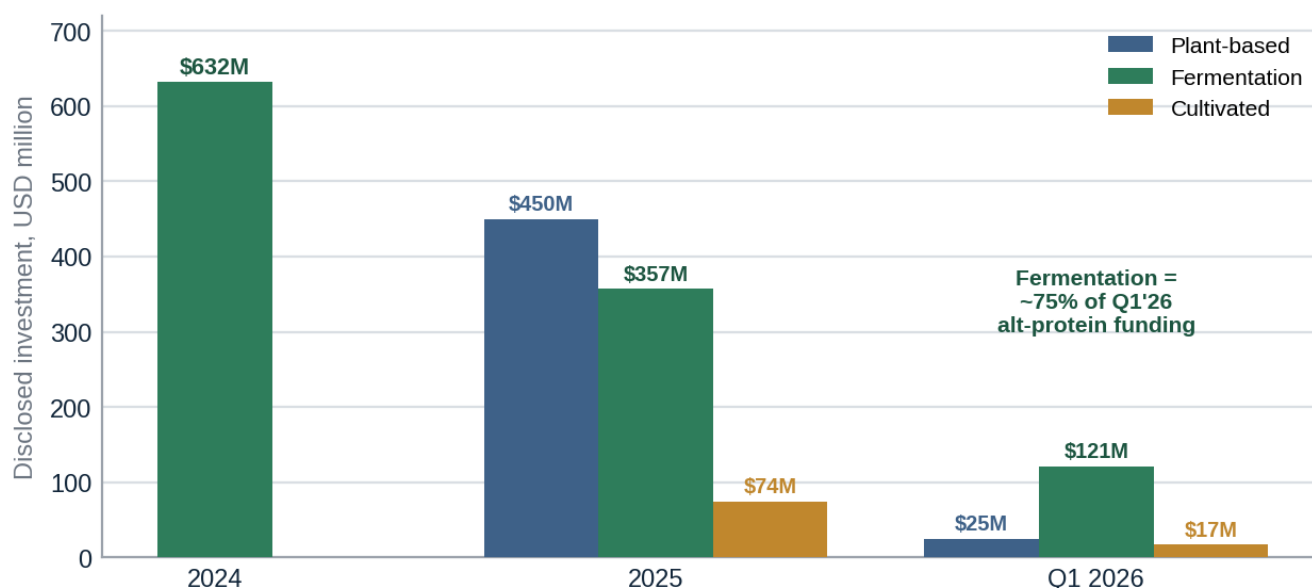
The capital picture

Money is voting for fermentation. Total alternative-protein funding was ~\$881M in 2025 (cumulative >\$19B since 2016). Within that, **fermentation now takes the lion's share of new capital — ~75% of a modest Q1 2026 total** — even as absolute fermentation funding cooled from its 2024 peak. The signal is a flight to the sub-sector with the clearest path to real production.

EXHIBIT 23

Capital is rotating into fermentation — on tougher terms

Disclosed alternative-protein investment by sub-sector.



Source: GFI analysis of Net Zero Insights data. 2024 shown for fermentation only.

Two things are true at once. The **direction** is favourable — fermentation is where sophisticated capital concentrates, and four of the five largest 2024 alt-protein rounds were fermentation companies. But the **terms** have hardened: the market has moved “from promise to proof,” and high-profile failures (Meati in biomass, Motif in precision, Arkeon in gas) mean investors now underwrite scale-up risk, offtake and unit economics — not just science.

Implication for a strategic entrant

A well-capitalised industrial operator is *advantaged* here. The companies dying are those dependent on fickle venture rounds and covenant-laden debt to fund capex. An entrant that can (a) self-fund a staged plant, (b) bring feedstock and offtake, and (c) take a 7–10-year view can acquire distressed assets and talent cheaply and build the low-cost position the venture-funded field cannot.

SECTION 13

Regulation as moat and map

Regulatory approval is a real barrier to entry — which makes it a moat once cleared. Routes differ sharply across the three target geographies, and sequencing them well can save years. The headline: **start where approval is fastest and cheapest, use it to build evidence, then tackle the slow-but-credible markets.**

EXHIBIT 24

A patchwork — sequence entry from easy to hard

Relative favourability of the regulatory pathway by geography.

	Pathway speed	Low cost / burden	Mycoprotein precedent	Market attractiveness
United States	Favorable	Favorable	Favorable	Favorable
Singapore	Favorable	Favorable	Favorable	Moderate
Brazil	Moderate	Moderate	Moderate	Favorable
Canada	Moderate	Moderate	Favorable	Moderate
European Union	Challenging	Challenging	Favorable	Favorable

Source: Regulatory registers and filings (FDA GRAS; EU Novel Food 2015/2283; Singapore SFA; Health Canada; ANVISA RDC 839/2023); independent analysis.

- **United States — fastest, but tightening.** GRAS has cleared Quorn, Nature's Fynd (2021), ENOUGH and Rhiza (2024). But the FDA has hardened its stance: it issued a **deficiency letter in February 2026** that forced The Protein Brewery to withdraw its Fermotein GRAS notice pending animal-study and antinutrient clarification. Self-affirmed GRAS is under political pressure — file early and build a full dossier.
- **Singapore — fast beachhead (approval, not scale).** A proactive novel-foods regime makes it an efficient first-approval and credibility market — but it is a small-volume market, and clearing SFA neither validates industrial cost nor reduces the FDA/EFSA work. Sequence by dossier transferability and option value, not headline speed alone.
- **Brazil — newly navigable.** ANVISA's RDC 839/2023 established a defined novel-foods pathway at end-2023 — a material improvement for a large, protein-hungry market with domestic feedstock advantages.
- **European Union — slow but credible, with a conditional reward.** The Novel Food Regulation is the world's most stringent (Fermotein took ~6 years), but Quorn is grandfathered (pre-1997) and mycoprotein is already approved. Note the data-protection reward is *not automatic*: under Art. 26 of Reg. 2015/2283, an applicant with qualifying newly-developed proprietary evidence **may** be granted 5-year protection on request — a useful barrier where earned, but it does not bar a competitor from filing an independent dossier.
- **United Kingdom — rising evidentiary bar.** FSA guidance (March 2026) now makes whole-genome sequencing and DIAAS digestibility testing effectively mandatory — more cost and time to budget for.

SECTION 14

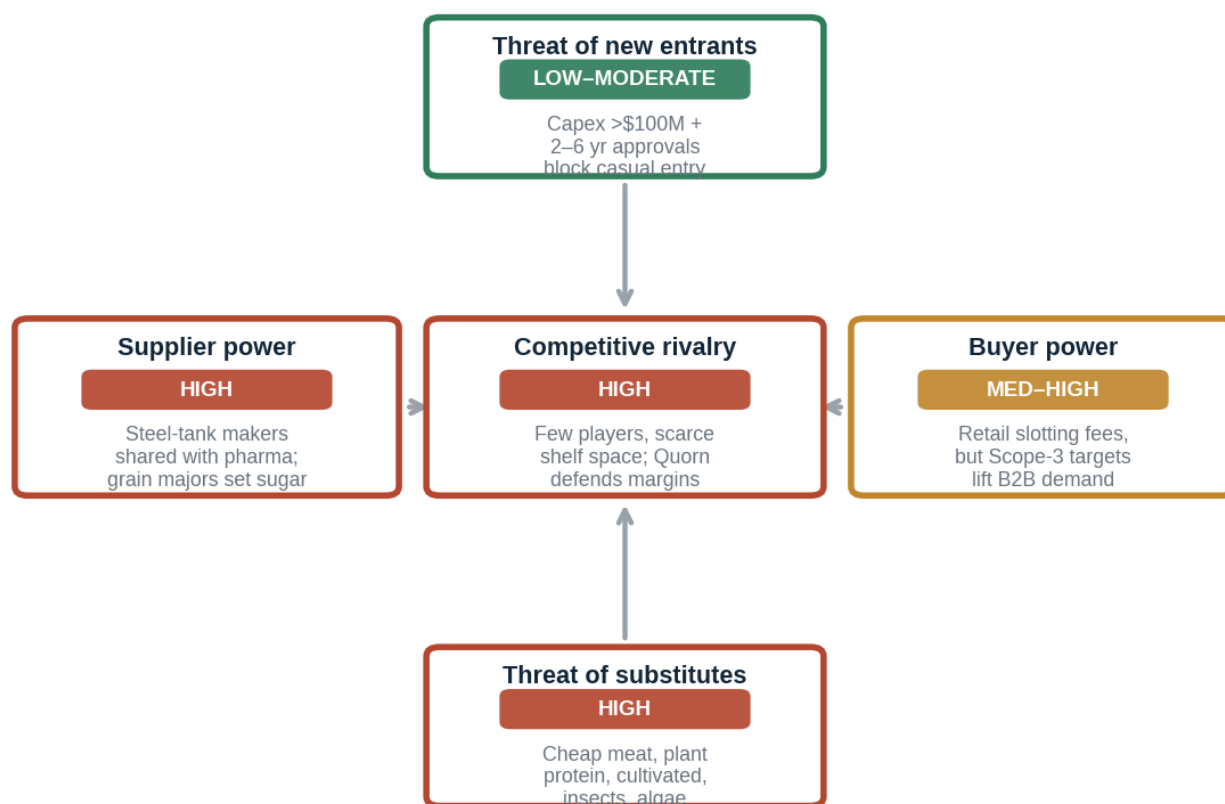
Industry attractiveness — Porter's Five Forces

Structurally, the mycoprotein industry is **moderately attractive and defensible for a well-capitalised incumbent**, precisely because the same forces that make entry hard (capital, regulation, supplier power) also protect those already inside.

EXHIBIT 25

The five forces favour a scaled, integrated entrant

Competitive-force intensities and their drivers.



Source: Independent analysis (Porter framework).

- **New entrants (low–moderate):** \$100M+ bioreactor capex and 2–6-year global approval labyrinths block trivial entry — good news for a serious entrant, bad news for casual ones.
- **Supplier power (high):** sterile steel-tank capacity is shared with Big Pharma (premium pricing), and glucose supply sits with grain-processing giants — an argument for feedstock integration.
- **Buyer power (medium–high):** retail slotting fees squeeze margins, but Fortune-500 Scope-3 urgency gives B2B suppliers leverage to lock in stable annual offtake.
- **Substitutes (high):** cheap subsidised meat, plant protein, and emerging cultivated/insect/algae platforms all compete for the same capital and attention.
- **Rivalry (high):** few players, scarce shelf space, and Quorn's gravitational pull force price discipline — reinforcing that cost leadership, not brand, is the winning axis.

SECTION 15

SWOT synthesis

The integrated view: mycoprotein's **strengths and opportunities are structural and durable** (sustainability, nutrition, GLP-1, hybrid, feedstock integration), while its **weaknesses and threats are largely executional and macro** (capex, RNA-yield drag, regulatory timing, perception) — which is exactly the risk profile a disciplined industrial operator is best placed to manage.

EXHIBIT 26

SWOT — durable upside, executional risk

STRENGTHS

- Organic fibrous texture without harsh chemical extrusion
- Best-in-class LCA — radical cuts in carbon, water, land
- Complete protein + prebiotic fibre; PDCAAS ~0.996
- Closed-system production, insulated from climate & zoonotic shocks

OPPORTUNITIES

- Capitalise on the GLP-1 & active-ageing megatrend (\$15–25/kg)
- Hybrid 50/50 formulations for QSR & foodservice
- Feedstock co-location + upcycling (ENOUGH–Cargill model)
- B2B diversification into industrial binders & materials

WEAKNESSES

- Overwhelming capex to reach genuine scale
- Unit-economics drag from RNA reduction (~30% biomass loss)
- Neutral base flavour needs formulation to mimic meat
- Strain-mutation cycles force costly production stoppages

THREATS

- Unpredictable FDA/EFSA timelines stalling cash flow
- Structural capital crunch felling under-funded rivals
- 'Ultra-processed' perception & consumer fatigue
- Continued decline in retail chicken prices

Source: Independent analysis.

SECTION 16

Scenario-based demand model & plant sizing

This is the study's core deliverable: a bottom-up estimate of **annual mycoprotein demand in tonnes/year** to serve as the sizing premise for the commercial fermentation plant. Rather than extrapolate a dollar-market CAGR, we build demand from physical volumes, applying a transparent **TAM** → **SAM** → **penetration** chain to each of two application scenarios, across Brazil, the United States and Europe, under conservative, moderate and aggressive cases.

Method — and the unit discipline that matters

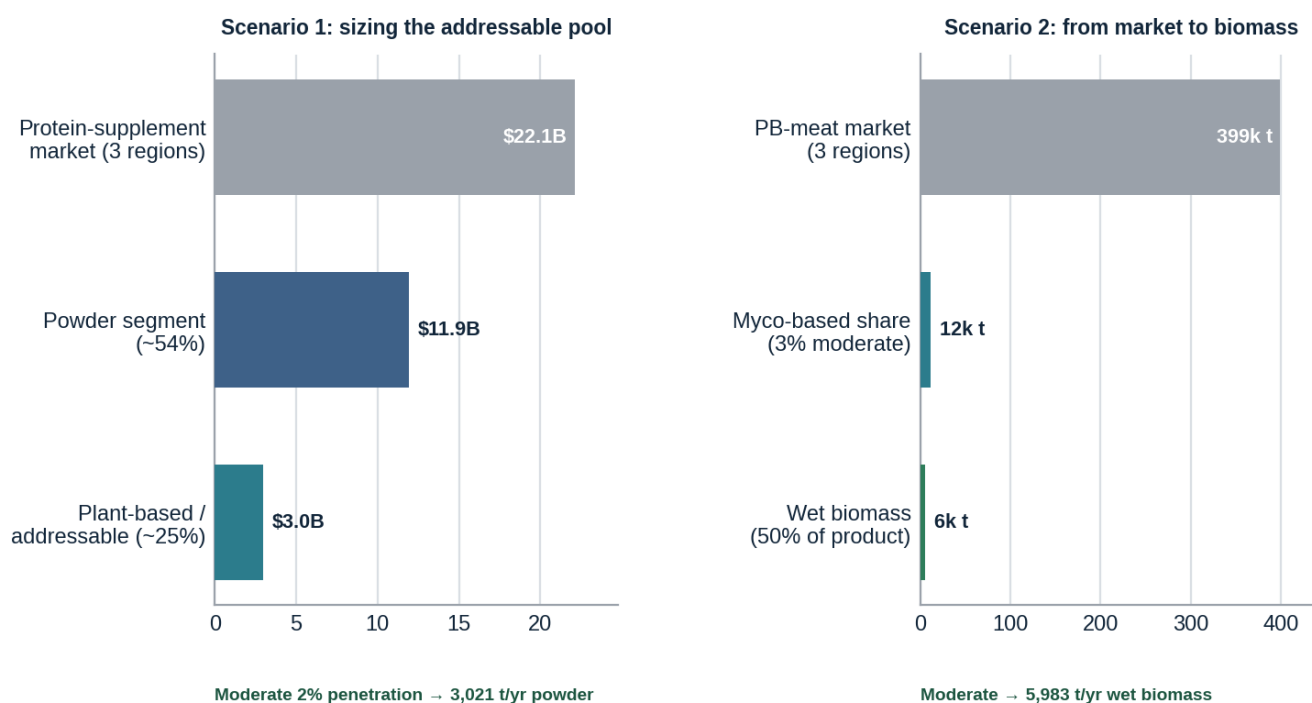
Every figure below is tagged to an explicit basis, because mixing them is the fastest way to a wrong answer.

Scenario 1 (supplement powder) is modelled in **tonnes of dry powder** (~90% solids, ~55% protein). **Scenario 2** (meat-analogue) is modelled in **tonnes of wet biomass** (paste at ~25% solids). To size one fermenter we convert both to a common **wet-biomass-equivalent** (1 t dry powder ≈ 3.6 t wet paste; a finished analogue is ~50% wet paste by mass). Addressable share (SAM) is the slice a novel fungal ingredient can realistically contest — not the whole protein market.

EXHIBIT 27

How the demand funnel is built (moderate case shown)

From total market to physical mycoprotein tonnes, for each scenario.



Source: Independent model; market sizes triangulated (protein-supplement market ~\$28–30B; plant-based meat per GFI/Circana scanner data and vendor reports). Illustrative — calibrate to an internal demand model.

Scenario 1 — mycoprotein as a human-supplement ingredient

Mycoprotein sold as a protein powder (alone or blended), competing with whey, soy isolate and pea protein on a *protein-plus-fibre* and sustainability proposition rather than on grams of protein per dollar — an important caveat, since at ~50–55% protein it is less dense than the 80–90% isolates.

- **Highest-potential markets:** the **US** is the largest supplement market by value (~40% of a ~\$28–30B global market) with the deepest sports-nutrition infrastructure; **Brazil** is a large, fast-growing, fitness-driven supplement market and JBS's natural home advantage; **Europe** is the fastest *route to market* because mycoprotein is already approved there. Sequencing: Europe (no regulatory wait) and Brazil (home base) first, US as the scale prize.
- **Market size & growth:** global protein supplements ~\$28–30B in 2025, growing ~6–10%/yr; powder is ~54% of that. The realistically addressable slice for a novel fungal protein is the **plant-based / alternative-protein powder** segment (~25% of powder and rising), i.e. ~\$2.9B and ~**150,000 t/yr** of product across the three regions.
- **Competitors & substitutes:** whey (Optimum Nutrition/Glanbia), soy isolate (ADM), pea isolate (Roquette, Puris), plus mycoprotein peers positioning powders (Marlow/Quorn ingredients, ENOUGH, The Protein Brewery). Whey is the entrenched default and the true benchmark.
- **Prices:** whey concentrate ~\$14.7/kg wholesale (~\$17/kg on a 100%-protein basis); retail whey ~\$25–32/kg; pea isolate ~\$15–25 retail (~\$5–9 bulk); soy isolate ~\$12–20 retail. A mycoprotein powder would sit at an estimated ~\$20–30/kg retail — defensible only on its fibre, completeness and sustainability, given lower protein density.
- **Realistic share:** as a novel ingredient facing entrenched incumbents and neophobia, a **0.5% (conservative) to 5% (aggressive)** penetration of the plant-protein-powder SAM — i.e. only ~0.1–1.3% of the *total* powder market — is a defensible band.

EXHIBIT 28**Scenario 1 — mycoprotein powder demand (dry basis)**

Region	Supplement mkt 2025	Addressable powder (t/yr)	Conserv. (t/yr)	Moderate (t/yr)	Aggress. (t/yr)
United States	~\$11.0B	~74,000	~370	~1,500	~3,700
Europe	~\$8.6B	~58,000	~290	~1,160	~2,900
Brazil	~\$2.5B	~19,000	~95	~375	~940
Total (powder)	—	~151,000	~750	~3,000	~7,600

Source: Independent model. Powder ≈ dry biomass; multiply by ~3.6 for wet-biomass equivalent.

Scenario 2 — mycoprotein as a meat-analogue ingredient

Wet biomass as the primary ingredient in meat-alternative products, competing with plant-based (soy/pea) analogues on texture and nutrition. Note the honest headwind: US retail plant-based meat has been *declining* (nearing 2019 levels) and European volume is roughly flat — so this scenario is a share-of-a-mature-category play, with hybrid meat (Section 4) as the larger but less-proven upside.

- **Highest-potential markets: Europe** is the most attractive near term — largest regional market, already-approved, and primed by Quorn familiarity; the **US** is the biggest single market but its retail segment is shrinking (approach via foodservice/hybrid, not branded retail); **Brazil** is smaller but growing ~11%/yr and is JBS's home turf with distribution reach (Seara, Incrível).
- **Market size & outlook:** defensibly ~\$1.5B (US, retail-scanner-anchored; vendor all-channel figures run \$3–4B), ~\$2.7B (Europe), ~\$0.3B (Brazil) — together ~**400,000 t/yr** of finished plant-based-meat product. Growth is contested: vendor CAGRs of 15–20% sit against flat-to-declining scanner volumes, so we treat the physical base as roughly stable, not booming.
- **Competitors:** Beyond Meat, Impossible, and soy/pea incumbents globally; in Brazil, Fazenda Futuro (Future Farm), Marfrig (PicNic), BRF (Sadia Veg) and JBS's own Seara Incrível; mycoprotein incumbent Quorn is the reference point.
- **Prices:** plant-based meat retails ~\$10–14/kg — roughly 2× beef and 4× chicken per kg (a ~77% premium in 2023, now narrowing). A mycoprotein-based product would target ~\$9–13/kg.
- **Realistic share:** a **1% (conservative) to 7% (aggressive)** share of the plant-based-meat market for mycoprotein-based products, reflecting genuine texture/nutrition advantages against entrenched, cheaper soy/pea incumbents.

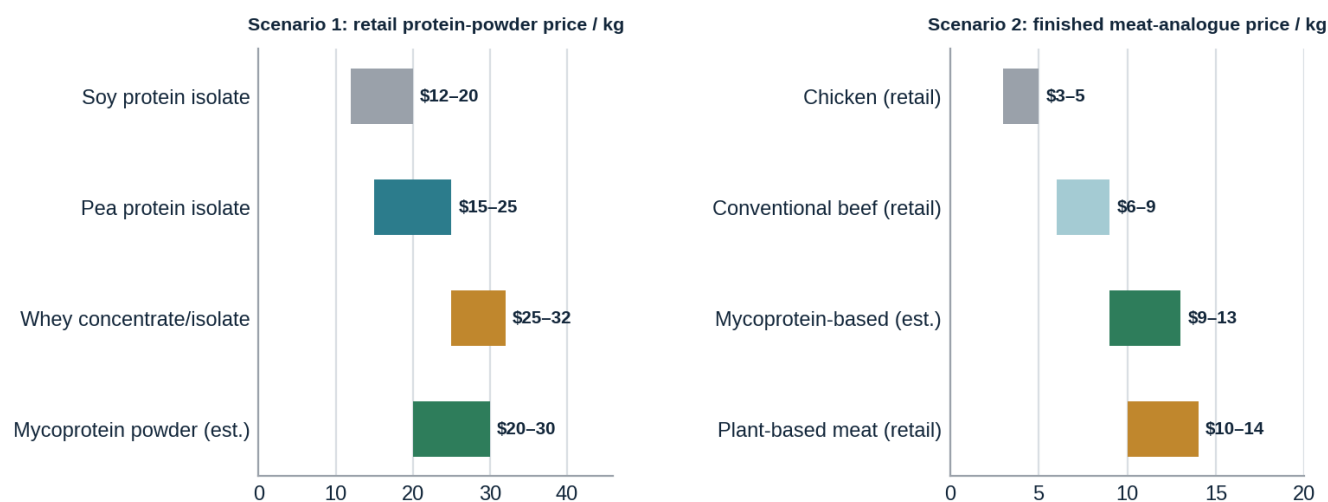
EXHIBIT 29**Scenario 2 — mycoprotein wet-biomass demand**

Region	PB-meat mkt 2025	Market volume (t/yr)	Conserv. (t/yr)	Moderate (t/yr)	Aggress. (t/yr)
United States	~\$1.5B	~136,000	~680	~2,050	~4,770
Europe	~\$2.7B	~225,000	~1,125	~3,375	~7,875
Brazil	~\$0.3B	~38,000	~190	~565	~1,315
Total (wet biomass)	—	~399,000	~2,000	~6,000	~14,000

Source: Independent model; wet biomass = myco-based product × ~50% paste content.

EXHIBIT 30**Reference prices for both scenarios**

Where mycoprotein would sit against the competitors it must displace.



Source: Green Circle Capital protein pricing review; GFI retail pricing; retail surveys.

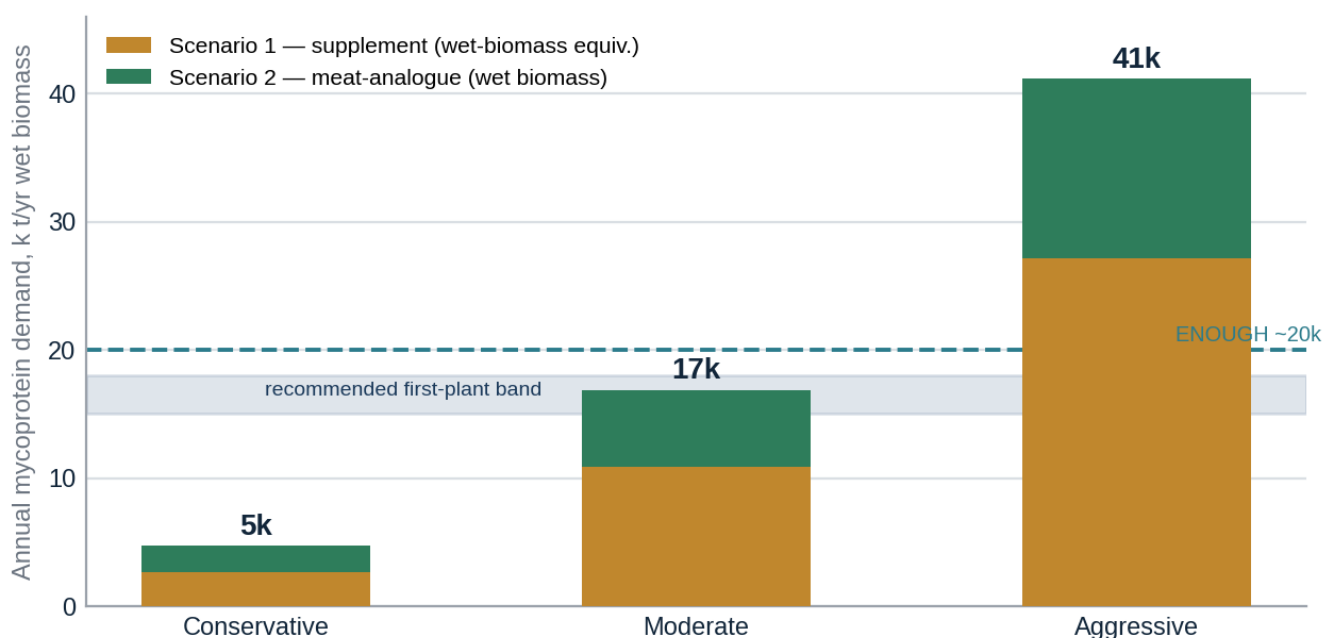
Synthesis — the annual-demand premise for plant sizing

Converting both scenarios to a common wet-biomass-equivalent and summing them gives the headline sizing premise. The **moderate case lands at ~17,000 t/yr of wet biomass — which independently derives, bottom-up, the ~15–20k first-plant band** that capacity benchmarks also suggest. The conservative case (~4,700 t/yr) argues for a de-risked first line, and the aggressive case (~41,000 t/yr) is a multi-year expansion target approaching twice ENOUGH's current scale.

EXHIBIT 31**Combined annual demand → plant nameplate**

Case	Scenario 1 (wet-equiv.)	Scenario 2 (wet)	Total wet biomass	~Dry biomass	Sizing read
Conservative	~2,700 t/yr	~2,000 t/yr	~4,700 t/yr	~1,200 t/yr	De-risked first line
Moderate	~10,900 t/yr	~6,000 t/yr	~16,900 t/yr	~4,200 t/yr	Recommended first plant
Aggressive	~27,200 t/yr	~14,000 t/yr	~41,100 t/yr	~10,300 t/yr	Long-term expansion

Source: Independent model. Wet-biomass equivalent is the fermenter-output basis for sizing.



Source: Independent model; ENOUGH benchmark from company disclosures.

The sizing recommendation — and its caveats

Design a first plant with a nameplate around **15,000–18,000 t/yr of wet biomass** (the moderate case), engineered for phased expansion toward ~30–40k as demand proves out, and consider commissioning an initial de-risked line nearer **5,000–8,000 t/yr** if secured offtake is thin at FID. Three caveats bind this: (1) the penetration rates are the model's most sensitive input — a 1-point shift moves the answer materially; (2) the plant-based-meat base is flat-to-declining, so Scenario 2 upside leans on hybrid and foodservice, not branded retail; (3) these are demand-side volumes — they set a *ceiling* for sizing, and the binding constraint remains securing offtake for a majority of nameplate before committing capital.

SECTION 17

Financial model, returns and sensitivity

A capacity recommendation is not complete until it clears a return hurdle. This section converts the demand volumes of Section 16 and the cost curve of Section 11 into **cash flows, NPV, IRR, payback and break-even conditions** for each candidate plant size. Every input is an explicit assumption listed in Appendix A and is meant to be overwritten with the desk's own feedstock, capex and price numbers — the structure is the deliverable here, not the specific figures.

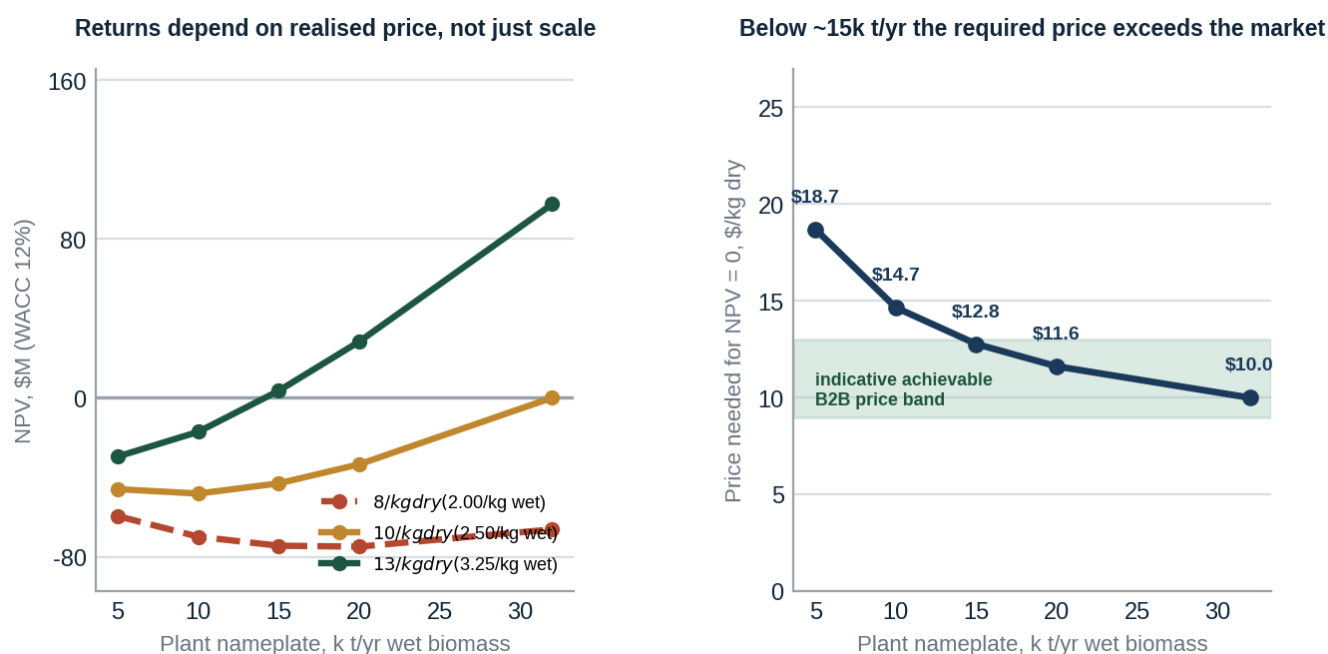
Read this section as a framework, not a valuation

The model uses a 15-year life, a 12% discount rate, a three-year commercial ramp (40% / 70% / 90% utilisation), working capital at 15% of revenue, and maintenance capex at 3% of gross capex. Capex scales with a 0.65 exponent from a \$120M anchor at 20k t/yr; fixed opex scales with a 0.5 exponent. **None of these are JBS figures.** Replace them before the numbers are used for anything other than shaping the questions.

EXHIBIT 32

Returns depend on realised price at least as much as on scale

Left: NPV by nameplate at three realised price levels. Right: the price each plant size must achieve to return the cost of capital.



Source: Independent discounted-cash-flow model; assumptions in Appendix A.

EXHIBIT 33

Returns by plant size at an assumed \$2.50/kg wet realised price

Nameplate (wet)	Capex	EBITDA at 90%	NPV @ \$2.50/kg wet	IRR	Price needed for NPV=0
5,000 t/yr	~\$49M	~\$3M/yr	~\$46M	negative	\$4.67/kg wet
10,000 t/yr	~\$76M	~\$9M/yr	~\$48M	~0%	\$3.66/kg wet
15,000 t/yr	~\$100M	~\$16M/yr	~\$43M	~4.5%	\$3.19/kg wet
20,000 t/yr	~\$120M	~\$24M/yr	~\$34M	~7.3%	\$2.90/kg wet
32,000 t/yr	~\$163M	~\$42M/yr	~\$0M	~12.0%	\$2.50/kg wet

Source: Independent DCF model. Negative NPVs at the assumed price are the point of the exhibit: the project is priced, not merely sized.

The single most important output in this report

At a realised price of **\$2.50/kg wet** — the figure implied by the commonly-quoted ~\$2.50/kg cost benchmark — **only the 32k t/yr plant returns the cost of capital**. A 15k plant needs to realise about **\$3.19/kg wet** (~\$12.77/kg dry-equivalent) to reach NPV zero. That price is plausible for a differentiated B2B ingredient, but it sits at the upper end of the achievable band — which means the investment case rests on **securing price, not just volume**. This is the quantitative reason the recommendation is *proceed to feasibility* rather than *invest*: binding offtake at a defined price is not a nice-to-have, it is the variable that decides the outcome.

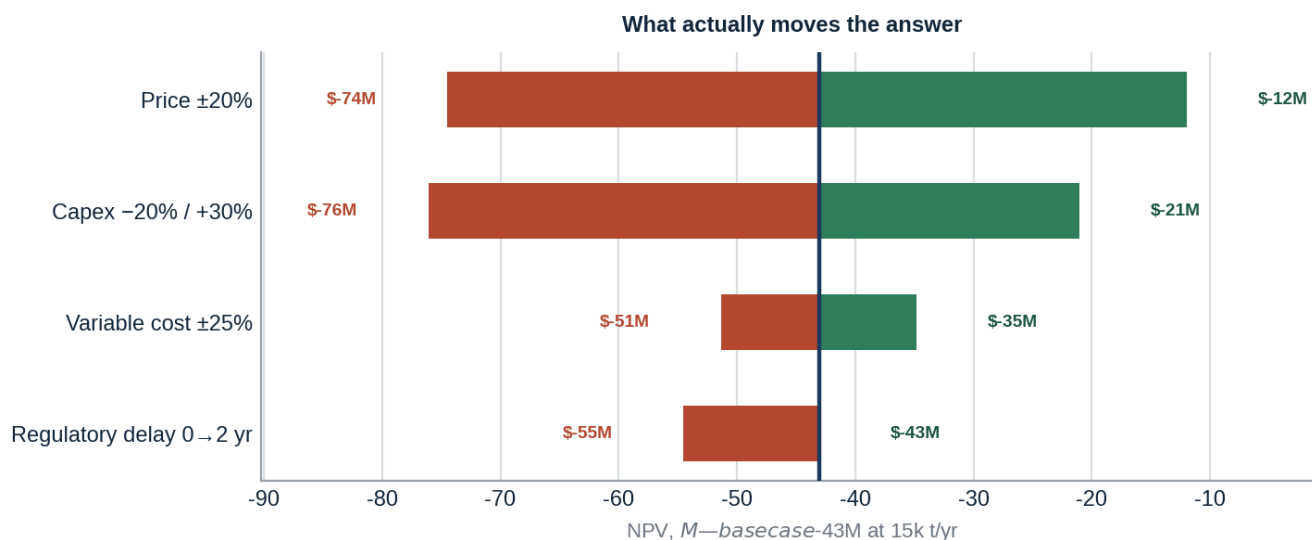
What actually moves the answer

Sensitivity run on the 15k t/yr case. Price and capital cost dominate; a two-year regulatory delay is painful but survivable; variable-cost improvement alone cannot rescue the case.

EXHIBIT 34

Price and capex dominate the outcome

NPV swing at 15k t/yr under single-variable stress.



Source: Independent DCF model, one variable moved at a time from the base case.

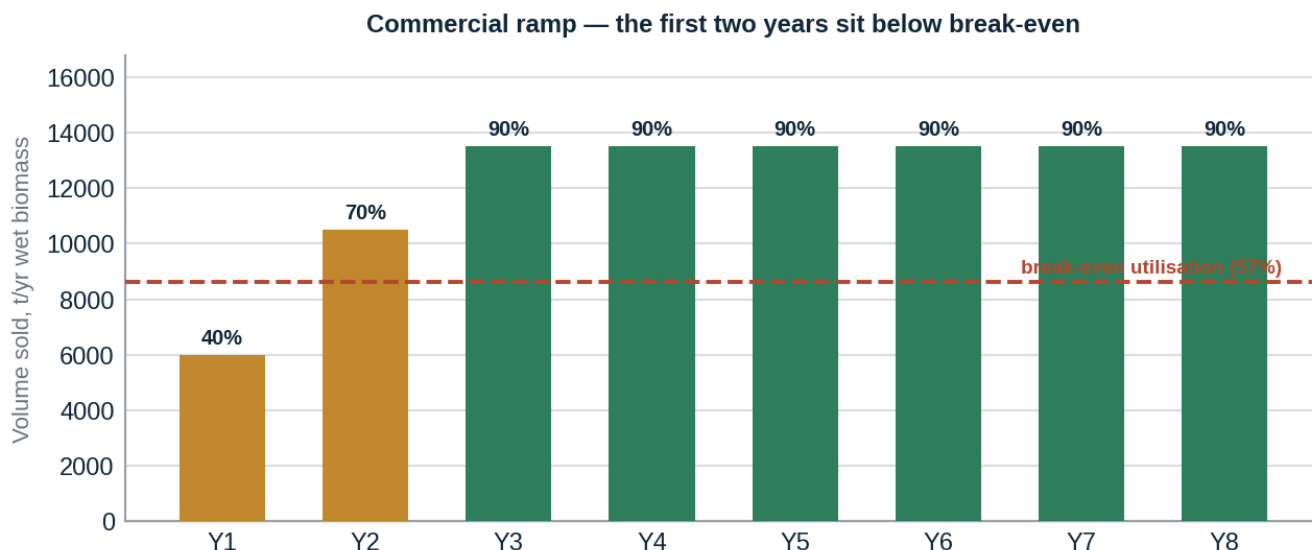
Commercial ramp and plant utilisation

Demand does not arrive on commissioning day. Applying a realistic ramp to the Section 16 volumes shows the first two years running *below* break-even utilisation — a funded, expected feature of the plan rather than a surprise, and a reason to size the balance sheet for it.

EXHIBIT 35

The first two years run below break-even by design

Volume sold against nameplate for a 15k t/yr plant on a three-year ramp.



Source: Independent model; ramp assumption 40% / 70% / 90% utilisation.

Three conditions before any final investment decision

1. Normalise all economics on a single stated basis (\$/kg dry, \$/kg protein and \$/kg finished product) including downstream processing, formulation, packaging and logistics — the framework is in Section 11 and needs the desk's real cost inputs. **2. Re-run this model** with JBS feedstock, capex and price data at 10k / 15k / 20k / 32k t/yr. **3. Secure binding offtake** — not letters of intent — covering **60–70% of first-line nameplate** at defined prices and specifications. On the evidence assembled here, condition 3 is the one most likely to decide whether this project should exist.

SECTION 18

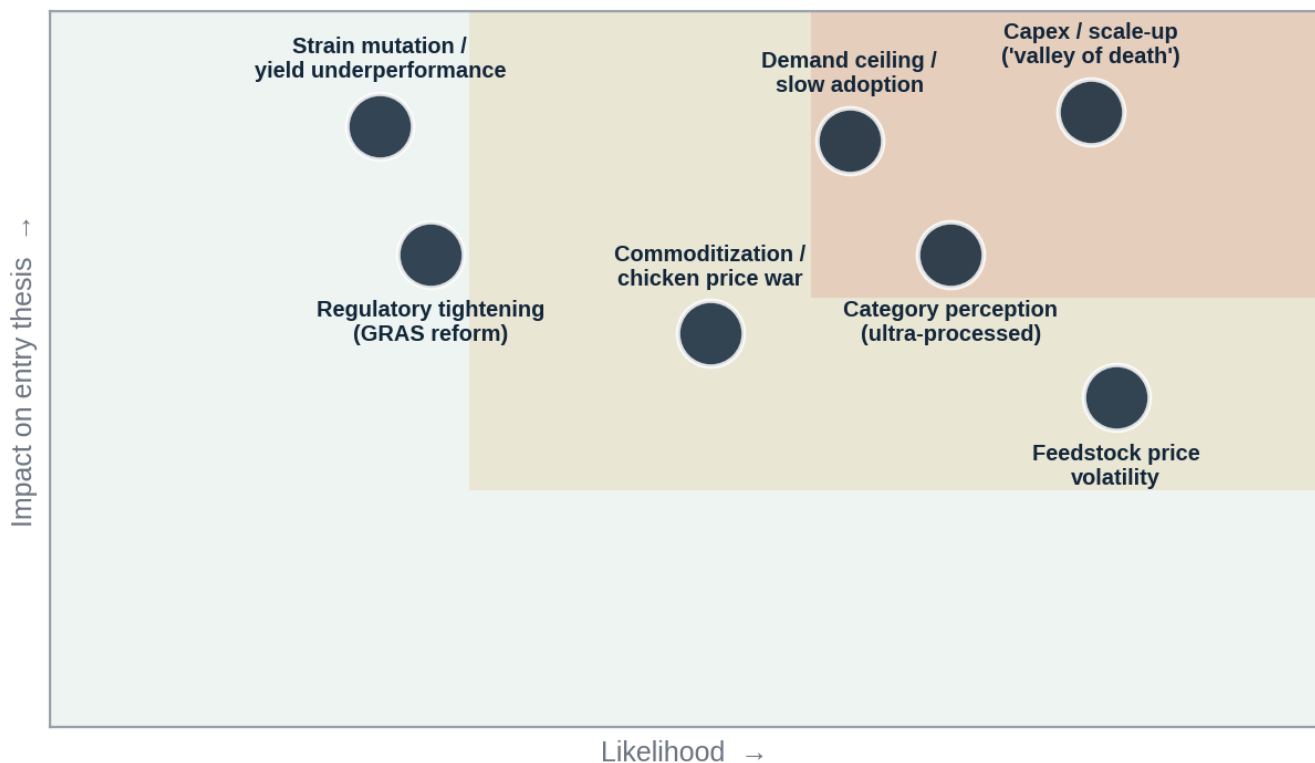
Risks and sensitivities

The dominant risks cluster in execution and demand — exactly where an experienced industrial operator with feedstock and offtake has the most control. The risks an entrant controls *least* (perception, regulatory reform) are real but second-order to getting the plant built and sold-out.

EXHIBIT 36

Execution and demand dominate the risk map

Key risks by likelihood and impact on the entry thesis.



Source: Independent analysis.

Mitigations

- **Capex / scale-up:** stage capacity; license a proven process rather than inventing one; partner with an operator who has already scaled a fermentation line.
- **Demand ceiling:** de-risk with binding offtake before construction; anchor to a large customer's reformulation programme; keep the plant flexible across applications and the GLP-1 niche.
- **Feedstock & RNA-yield:** secure long-term carbohydrate supply and co-locate; invest R&D in low-RNA, thermotolerant strains to attack the single biggest cost/yield drag.
- **Strain mutation:** treat continuous-culture stability as core IP; budget planned stop/sterilise/re-inoculate cycles into opex.
- **Perception:** lean on the whole-food, minimal-processing, high-fibre story; favour hybrid and foodservice channels less exposed to front-of-pack scrutiny.
- **Regulatory:** file early under current rules; diversify approvals across geographies so no single regime is a chokepoint.

SECTION 19

Strategic options and the recommendation

Three business models are on the table, and the evidence strongly favours one on a risk-adjusted basis in the current cycle.

- **Vertically-integrated B2C (highest risk now):** build the plant, build the brand, and educate the consumer through marketing — the capital-heavy path Meati's collapse illustrates (though that failure was partly a covenant/liquidity event, the vulnerability itself flowed from the model). Quorn is the lone survivor here, cushioned by a parent.
- **B2B ingredient supplier (recommended core):** ENOUGH and The Protein Brewery are the model — decouple from marketing, optimise volumetric yield, and use large offtake agreements as financial collateral for debt. ENOUGH's Cargill co-location dramatically cut logistics and feedstock capex.
- **Hybrid-meat formulator (tactical expansion):** blend 30–50% mycelium into conventional meat to cut emissions and add fibre without alienating meat-eaters; QuornPro's UltiMeat foodservice deal validated it in 2025/26.

EXHIBIT 37

Build vs partner vs buy — partner-then-build wins

Route	Speed	Capex exposure	Control / IP	Key risk	Fit for entrant
Build (cold greenfield)	Slow	Highest	Full	Scale-up & demand risk borne alone	Weak alone
Partner / offtake / JV	Fast	Shared / staged	Shared	Partner dependency	Recommended
Buy (distressed assets)	Fast	Low (cents on \$)	High	Integration & tech fit	Accelerant

Source: Independent analysis.

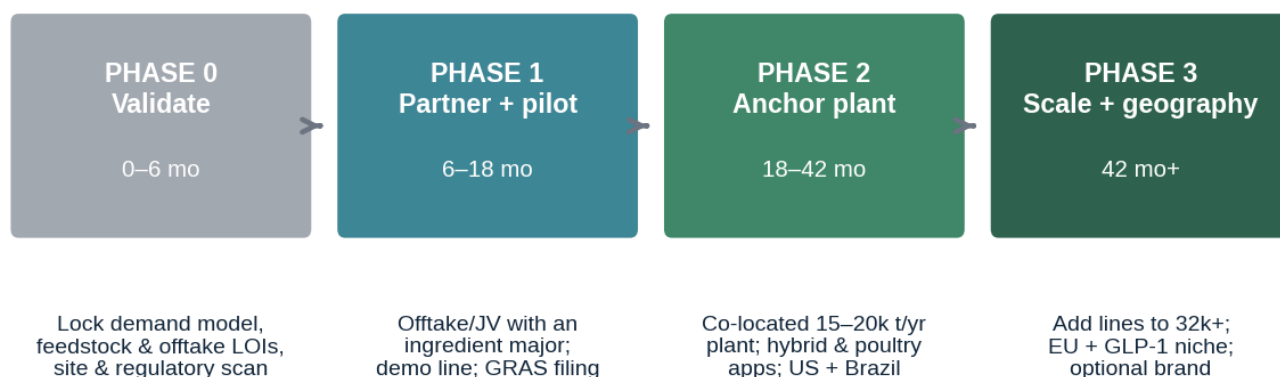
Where to play / how to win — in one line each

- **Where to play:** B2B mycoprotein ingredient supply at commodity scale (Scenario 2), plus the higher-margin active-nutrition/supplement segment (Scenario 1) and GLP-1 as an option-value adjacency — rather than a capital-intensive branded-retail meat business.
- **How to win:** lowest delivered cost (feedstock co-location + staged capex + low-RNA strains) and best application performance (hybrid and poultry-style first).
- **Where to build:** co-located with carbohydrate feedstock; serve already-approved Europe and home-market Brazil first, with the US as the scale prize once its retail segment stabilises.
- **How big:** a first plant of ~15–18k t/yr wet biomass (the Section 16 moderate case), staged from a smaller de-risked line and expandable toward ~30–40k — sized to binding offtake, not to the demand ceiling.

A phased playbook

EXHIBIT 38

A four-phase entry playbook — validate, partner, anchor, scale



Source: Independent analysis.

The demand premise, restated

Size the conceptual plant to ~15,000–18,000 t/yr of wet biomass — the moderate demand case of Section 16 — engineered for phased expansion toward ~30–40k t/yr as demand proves out. Do not size below ~10k t/yr for a commercial asset: the cost curve in Section 11 shows a 5k line needs 93% utilisation merely to break even. Treat the demand estimate as a *ceiling* that must be converted into contracted volume: the returns analysis in Section 17 shows a 15k plant needs to realise about **\$3.19/kg wet** to return its cost of capital, so **binding offtake covering 60–70% of first-line nameplate at defined prices** is the condition that decides the project. Commercially, the volume is most accessible as a **B2B ingredient and hybrid** business in already-approved Europe and home-market Brazil first, with the US as the scale prize — rather than as a capital-intensive branded retail entry.

SECTION A

Methodology, assumptions and sources

Approach

Figures were triangulated across independent categories of evidence, cross-checked against a parallel independent analysis, and reconciled to reasoned base cases. Market-size ranges reflect the spread of published third-party estimates; where those diverge materially, we show the range and a central case. Capacity, funding and financial figures rely on primary company disclosures and filings where available. The Section 16 demand model is built bottom-up (market volume × addressable share × penetration, on explicit dry/wet/product bases) and remains illustrative — to be calibrated against the investor's own demand model and binding offtake before any capital commitment.

Key assumptions

- Mycoprotein 2025 market ~\$0.76B (finished product) to ~\$1.4B (broad ingredient); base-case growth ~9–10% (published range 5–17%).
- Biomass cost of production ~\$4–6/kg (basis-dependent), falling to ~\$2.5/kg at ~32k t/yr; greenfield capex ~\$80–150M for a 10–20k-tonne plant, i.e. ~\$4,000–15,000 per annual tonne on those figures.
- First-plant sizing band 15–18k t/yr, benchmarked to ENOUGH (Sas van Gent: ~10k t/yr at 2022 opening, doubled to ~20k with a second line, site engineered toward ~50–60k) and Quorn (~67k est.).
- LCA values are cradle-to-gate and vary by system boundary; beef GHG spans a wide 10–37 kg CO₂e/kg, and the advantage-vs-beef multiple varies accordingly.
- RNA-reduction biomass loss ~30%; strain-mutation cycle ~100 generations (~6 weeks) in continuous culture; continuous vs batch productivity up to ~5×.

Model assumptions — Sections 11, 16 and 17

The three quantitative models in this report are bottom-up and fully specified below so that every output can be reproduced or challenged. None of these values are JBS figures; all are intended to be replaced with desk data.

EXHIBIT 39

Complete model assumptions

Parameter	Value used	Basis / note
Wet paste solids	25%	Fermenter output basis for sizing
Protein in dry biomass	50%	Range 45–55%; strain-specific
Dry powder solids	90%	Supplement-powder form
Paste in finished analogue	50%	Mass fraction of myco-based product
Glucose price	\$0.40/kg	TEA viability threshold
Biomass yield on glucose	0.45 kg/kg	Pre-RNA-reduction
RNA-reduction solids loss	25%	Report range up to ~30%
Variable cost	\$2.09/kg dry	Feedstock + nutrients + utilities + consumables
Capex anchor / exponent	\$120M @20k / 0.65	Capacity-scaling rule
Fixed opex anchor / exponent	\$12M @20k / 0.50	Labour, maintenance, overhead
Plant life / depreciation	15 yr straight-line	Also the DCF horizon
Discount rate (WACC)	12%	Placeholder — replace with JBS hurdle

Parameter	Value used	Basis / note
Commercial ramp	40% / 70% / 90%	Years 1–3, then 90% steady
Working capital	15% of revenue	Released at end of horizon
Maintenance capex	3% of gross capex/yr	Sustaining capital
Tax rate	25%	Placeholder
S1 penetration (cons./mod./agg.)	0.5% / 2% / 5%	Of plant-protein-powder SAM
S2 share (cons./mod./agg.)	1% / 3% / 7%	Of plant-based-meat market

Source: Independent models. Every row is an overwritable input.

Principal sources

The Good Food Institute (State-of-the-Industry and investment analyses, drawing on Net Zero Insights); SPINS and Euromonitor retail/consumption data; company disclosures, annual accounts and press (Monde Nissin/Marlow Foods, ENOUGH, Nature's Fynd, Emergy/Meati, The Protein Brewery, Enifer, The Better Meat Co, Cargill); peer-reviewed life-cycle assessments, techno-economic models and bioprocess-engineering references (including work on mycoprotein RNA reduction and continuous fermentation); market-research syntheses (Precedence, FMI, Fact.MR, DataM, Persistence, IndexBox and others); and regulatory registers (US FDA GRAS notices incl. GRN 1268; EU Novel Food Regulation 2015/2283 and EFSA opinions; Singapore Food Agency; Health Canada; Brazil ANVISA RDC 839/2023; UK FSA guidance). Specific figures are directional and current to mid-2026.

Important notice

This document was prepared as an independent strategic analysis for discussion purposes. It is not investment, legal, tax, accounting or medical/nutritional advice, and it does not constitute an offer or recommendation to buy or sell any security or asset. Market figures are estimates triangulated from third-party sources believed to be reliable but not independently verified; actual results may differ materially. Any investment decision should rely on independent due diligence and professional advice. For health, physiological or GLP-1-therapy questions, consult a qualified medical professional.